



Grad-CAM (Gradient-weighted Class Activation Mapping): A systematic literature review

Abdul Muiz Fayyaz ^{a,*,} Said Jadid Abdulkadir ^{a,b,}*, Noureen Talpur ^{a,b,},
Safwan Mahmood Al-Selwi ^{a,b,}, Shahab Ul Hassan ^{a,c,}, Ebrahim Hamid Sumiea ^a

^a Department of Computing, Universiti Teknologi PETRONAS, Seri Iskandar, 32610, Perak, Malaysia

^b Center for Research in Data Science (CeRDs), Universiti Teknologi PETRONAS, Seri Iskandar, 32610, Perak, Malaysia

^c Centre for Intelligent Signal & Imaging Research (CISIR), Universiti Teknologi PETRONAS, Seri Iskandar, 32610, Perak, Malaysia

ARTICLE INFO

Dataset link: <https://github.com/abdulmuizfayyaz/GradCAM-SLR>

Keywords:

Artificial Intelligence
Deep Learning
Explainable Artificial Intelligence
Grad-CAM
Systematic Literature Review

ABSTRACT

Explainable Artificial Intelligence (XAI) has become a crucial aspect of modern Machine Learning (ML) and Deep Learning (DL) applications, emphasizing transparency and trust in model predictions. Among various XAI techniques, Gradient-weighted Class Activation Mapping (Grad-CAM) stands out for its ability to visually interpret Convolutional Neural Networks (CNNs) by highlighting image regions that contribute significantly to decision-making. This Systematic Literature Review (SLR) provides a comprehensive analysis of Grad-CAM, its advancements in medical imaging, and applications in ML and DL. The review explores current research trends, variations of Grad-CAM, and its integration with different ML/DL architectures. A systematic search across Scopus, Web of Science, IEEE Xplore, and ScienceDirect identified 427 peer-reviewed publications (2020–2024), of which 51 were selected for in-depth examination. This study offers valuable insights into the evolution of Grad-CAM, its optimization techniques, and its role in improving model interpretability in medical imaging analysis and related fields.

1. Introduction

Explainable Artificial Intelligence (XAI) [1,2] is a technique for developing machine learning (ML) [3] and artificial intelligence (AI) [4] models and systems in a way that makes it possible for humans to understand and interpret their actions [5,6]. In AI, ML, and Deep Learning (DL), algorithms are used to reach a result [7], but the architects of these systems do not fully comprehend how they came to those results [8]. AI models and their inner workings are intended to be better understood by humans. This includes describing how the model values the components and how it makes predictions and conclusions based on those components. By emphasizing model architecture, data sources, and decision-making processes, XAI enhances transparency in AI systems, making the AI's behavior easier for users to trust and validate. It is simpler to hold AI systems responsible for their actions when they are explicable. This is crucial in fields like healthcare [9], and autonomous cars [10,11] where decisions can have significant consequences. By enabling people to understand how decisions are impacted by various circumstances, XAI can aid in detecting and reducing the effects of bias in AI systems.

XAI makes effective cooperation between humans and AI systems possible. Users may interact with AI more productively, identifying faults and adjusting its behavior [12]. For accountability and transparency in AI systems, there are legal and regulatory obligations in specific fields. The healthcare [9] industry has been increasingly adopting AI in recent years to improve disease diagnosis and reduce health risks. However, the development of trustworthy and XAI in healthcare is still in its early stages. This review aims to assess the trustworthiness and explainability of AI applications in healthcare by addressing concerns about methodological quality and bias risk. The study identified 51 recent contributions from various databases and categorized them into seven areas, which include explainable robotics, prediction, decision support, blockchain, transparency, digital health, and review.

An important area of study known as XAI has been developed as a consequence of the increased demand for transparency and the complexities of modern ML algorithms. As algorithms get complex, they become “black boxes,” making it difficult for users as well as developers to properly understand their decision-making processes [13]. This lack of transparency raises serious issues, especially in sensitive industries

* Corresponding authors.

E-mail addresses: abdulmuizfayyaz@gmail.com (A.M. Fayyaz), saidjadid.a@utp.edu.my (S.J. Abdulkadir), noureen.talpur@utp.edu.my (N. Talpur), safwan_21002827@utp.edu.my (S.M. Al-Selwi), shahab_22009928@utp.edu.my (S.U. Hassan), ebrahim_22006040@utp.edu.my (E.H. Sumiea).

<https://doi.org/10.1016/j.combiomed.2025.111200>

Received 7 March 2025; Received in revised form 19 August 2025; Accepted 7 October 2025

Available online 17 October 2025

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Nomenclature

AI	Artificial Intelligence
AUC	Area Under Curve
CAM	Class Activation Mapping
CHS-Net	COVID-19 Hierarchical Segmentation Network
CNN	Convolutional Neural Network
CXR	Chest X-ray
DCNN	Deep Convolutional Neural Network
DenseNet	Densely Connected Convolutional Network
DenseNet-121	Densely Connected Network 121-layer model
DenseNet-169	Densely Connected Network 169-layer model
DenseNet-201	Densely Connected Network 201-layer model
DL	Deep Learning
DLM	Deep Learning Models
GoogLeNet	Google's Neural Network
Grad-CAM	Gradient-weighted Class Activation Mapping
GRIPS	Grad-CAM-based RISE (Randomized Input Sampling for Explanation)
HayCAM	Hierarchical Attention-based Class Activation Mapping
Inception	Inception CNN Model
Inception-v3	Inception Version 3
LightGBM	Light Gradient Boosting Machine
LIME	Local Interpretable Model-agnostic Explanations
LRP	Layer-wise Relevance Propagation
Mask R-CNN	Mask Region-Based Convolutional Neural Network
ML	Machine Learning
MobileNet	Mobile Neural Network
MobileNet-V2	MobileNet Version 2
MRI	Magnetic Resonance Imaging
NLP	Natural Language Processing
PCA	Principal Component Analysis
PReLU	Parametric Rectified Linear Unit
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
REDCap	Research Electronic Data Capture
ReLU	Rectified Linear Unit
Res2Net	Second Generation Residual Network
ResNet	Residual Network
ResNet-101	Residual Network 101-layer model
ResNet-50	Residual Network 50-layer model
SeCAM	Semantic Class Activation Mapping
SHAP	SHapley Additive exPlanations
SLR	Systematic Literature Review
TS-CNN	Time Series Convolutional Neural Network
VGG	Visual Geometry Group
VGG16	Visual Geometry Group 16-layer model
VGG19	Visual Geometry Group 19-layer model
ViT	Vision Transformer Model
XAI	Explainable Artificial Intelligence
Xception	Extreme Inception

XG-boost	eXtreme Gradient Boosting
YOLO	You Only Look Once

like healthcare and banking, where AI choices can have a big impact on human beings. One of the principal motivations for the advancement of XAI is to enhance the interpretability of AI models [14]. Interpretability is defined as the ability to comprehend and elucidate the rationale underlying an AI system's decision-making or predictive outcomes [15].

The focus of XAI research is on resolving the issues of interpretability and transparency that are caused by advanced ML models [16]. The development of techniques makes it possible for meaningful explanation of AI decisions [17]. Many sophisticated ML models, such as deep neural networks, are inherently difficult to comprehend because of their intricate and non-linear structure, in contrast to conventional rule-based systems [18]. XAI approaches seek to close this gap and promote responsibility and confidence by explaining AI model outputs that are intelligible by humans. XAI includes a wide variety of techniques and strategies. Some methods concentrate on producing post-hoc explanations, which are explanations that are produced after a prediction has been made by the model [19]. Examples are perturbation-based approaches, which evaluate how changes in input data impact model outputs, and feature significance techniques, which emphasize the most important elements in a decision [20]. On the other hand, models that are inherently interpretable, like decision trees or linear models, are aimed to be clear from the start and have explainability built into them.

Gradient-weighted Class Activation Mapping (Grad-CAM) [21] is of significant importance for the visualization and interpretation of the decision-making processes of convolutional neural networks (CNNs) [22] within the context of image-related applications. This capacity for visual interpretation is paramount in fields such as medical imaging, where the identification of particular regions of interest is critical for establishing confidence in the model's decision-making framework. Conversely, model-specific methodologies, which are customized to the distinct attributes of a specific algorithm, provide a more profound comprehension of the internal mechanisms that influence predictions. Nonetheless, alternative approaches, such as feature importance or coefficient analysis, may exhibit deficiencies in their ability to capture the intricate relationships present within complex models. GENESIM's [23] computational expense and potential lack of scalability might be a concern, especially in large datasets. Decision trees, while interpretable, might struggle with representing non-linear relationships effectively. Patho-GAN's [24] reliance on a generative adversarial network framework may introduce complexity that could be unnecessary for simpler models. In summary, the choice of Grad-CAM is grounded in its efficacy for visual interpretability and detailed insights into specific model architectures, respectively.

The decision is motivated by the nature of the data, the interpretability requirements, and the complexity of the models under consideration, providing a balanced approach for enhancing the transparency of the AI system. Grad-CAM is chosen for interpreting DL image classification models because it provides transparency, visual insights into the decision-making process of deep neural networks and spatial localization of important features in the input image, particularly effective for CNN architectures but still Grad-CAM have some limitations known as ambiguity, non-linearity, uncertainty, black-box problems, inconsistency, high dimension data and feature space problems. Grad-CAM represents a methodological approach employed for the visualization and interpretation of the decision-making processes inherent in deep neural networks, particularly within the framework of CNNs. The Grad-CAM serves to accentuate the segments of an input image that exert the most significant influence on the predictions rendered by the model [23–25].

The key contributions of this systematic literature review (SLR) are as follows:

- Introducing a comprehensive seven-step approach to incorporate research on the initial implementation and work of Grad-CAM (Shown in Fig. 5).
- Presenting a detailed review of several enhanced Grad-CAM-related methods designed only for more accurate XAI interpretation. These approaches include CAM, Grad-CAM++, Guided Grad-CAM, Eigen-CAM, Mask Grad-CAM, and HayCAM (Section 5.1).
- This study provides a comprehensive summary of diverse ML and DL techniques integrated with Grad-CAM to enhance model interpretability. The discussion focuses on these techniques individually, highlighting their relevance and applicability in image classification (Section 5.2).
- Analysis of the most often used performance metrics in the ML/DL methods utilizing Grad-CAM in classification tasks (Section 5.3).
- Explaining and highlighting significant findings through the synthesis of data using an accurate mapping process to demonstrate the research intensity in the specified subject area (Section 5.4).

The current analysis provides a comprehensive SLR of the existing literature, extending research from 2020 to 2024 and highlighting the following key contributions as discussed above. A systematic approach has been created to conduct a rigorous and comprehensive search informed by prior research. It consists of seven distinct stages as depicted in Fig. 5. The fundamental objective of this research is to highlight the uses of Grad-CAM. The major goal of this research is to examine Grad-CAM's gaps and concerns.

The rest of this paper is as follows: Section 2 includes the background about XAI and related studies. Section 3 contains the Grad-CAM background. Section 4 includes the utilized methodology in this SLR. Section 5 includes the synthesis of data and analysis of the extracted results. Finally, Section 6 concludes this paper.

2. Background

AI systems [26] have reached human efficiency in several tasks that used to be computationally impossible during the decade that followed, particularly after 2012 [27]. Advances in the subject have been made possible by an increase in accessible knowledge, as well as significant hardware advances paired with novel optimization methods [28,29]. These breakthroughs may also be attributed to high-quality open-source libraries, which enabled developers and researchers to swiftly write and test models. Improvements in voice recognition [30], image classification [31], object identification [32], classic (board) games, and many other areas have taken place in their development and diffusion to real-world applications [33]. The models of ML, a subtype of AI that relies on mathematical models to improve machine intelligence, are built and trained on given datasets to make predictions for any test samples without having to be explicitly programmed to do so [34,35]. DL is the most popular subset of ML at the moment, as it replicates how the human brain analyzes data and patterns to reach an outcome [11,36].

The rest of this section includes: Section 2.1 provides a brief overview of XAI. Section 2.2 compared our work with the previous works from other reviews, SLRs, and survey publications. Section 2.3 discuss the Application of XAI in healthcare imaging.

2.1. Explainable Artificial Intelligence (XAI)

This section provides a brief overview of XAI techniques used in DL for medical image analysis. The term “XAI” describes artificial neural networks that can provide end users with intelligible insights on how they function [37]. As this discipline is still in its infancy, there is currently no consensus over the exact definition of explainability in this context [38]. This includes any method that aims to provide an explanation for the model's prediction by analyzing medical images. XAI seeks to provide trustworthy and open AI techniques that can be included in

user decision-making. Even though XAI is a relatively new field, the need for clear and intelligible AI systems has led to a lot of interest in it in recent years. XAI has a taxonomy that systematically classifies various methods and approaches used to achieve interpretability and transparency in ML models. This taxonomy can be divided into different dimensions as shown in Fig. 1, each addressing specific aspects of the XAI landscape.

The diagnosis of medical conditions has been significantly transformed by DL, permitting the development of accurate and reliable computer-aided diagnostic instruments for conditions such as intestinal bleeding, cancer, breast cancer, and fractures [39]. Despite its disruptive effect, the healthcare business has several obstacles, including a shortage of balanced, annotated medical data, ethical privacy issues, susceptibility to adversarial assaults from concealed medical pictures, and problems with user and patient trust. These challenges are examined by Dhar et al. [40], who focus on social trust challenges in autonomous intelligent systems and how AI autonomy in healthcare may handle them.

XAI endeavors to enhance the comprehensibility and transparency of ML/DL models [41]. The methodologies associated with XAI facilitate the interpretability of ML models by converting abstract concepts acquired by computational processes into a framework that is accessible to human understanding. A prevalent technique within XAI involves the assessment of feature attributions, which are scores assigned to each input of the model that quantify their significance in relation to the model's predictions. Consequently, by identifying the features that influence the model's outputs, the results generated by XAI can be construed as elucidations for the forecasts rendered. While XAI has predominantly been employed to scrutinize the decision-making frameworks leveraged by ML models, it has also demonstrated efficacy as a pedagogical instrument [42].

The goal of XAI is to make AI models and their decision-making processes more understandable to users, ensuring transparency as shown in Fig. 2. By providing clear explanations of how AI reaches its conclusions, XAI helps build user trust and confidence. It also promotes accountability by making AI systems responsible for their actions and reducing bias by identifying and correcting unfair influences in decision-making. Additionally, XAI enables better collaboration between humans and AI, allowing users to interact more effectively with these systems. It helps organizations comply with legal and ethical standards for AI transparency and fairness. Researchers and developers can use XAI to detect errors, refine models, and improve AI performance. In critical fields such as healthcare, finance, and autonomous systems, XAI enhances safety by ensuring AI operates reliably. It also provides better decision support for professionals like doctors and financial analysts, helping them make informed choices. Lastly, XAI plays a key role in educating users about AI-driven recommendations and predictions, allowing them to understand and trust AI-assisted decision-making.

2.2. Related studies

This section compares our work with pertinent studies from other reviews, SLRs, and survey publications. These days, the two most popular research areas are healthcare and XAI [43]. Because researchers are actively engaged in these fields, XAI for healthcare and medicine includes a variety of questionnaires. One of the hardest things for researchers to do is to map highly distributed, diverse medical information across arbitrary high-dimensional areas. Explainable machine judgments would make it easier to manage the diverse data for relevant outcomes [44]. With explainable machine learning (XML), these problems can be fixed, and ML conclusions can become more transparent.

Despite our knowledge, no study or survey to date provides a comprehensive overview of metrics, datasets, medical imaging analysis using XML/DL, frequency of publication in the domain related to Grad-CAM, and its variation. Adadi and Berrada [45] looked into XAI

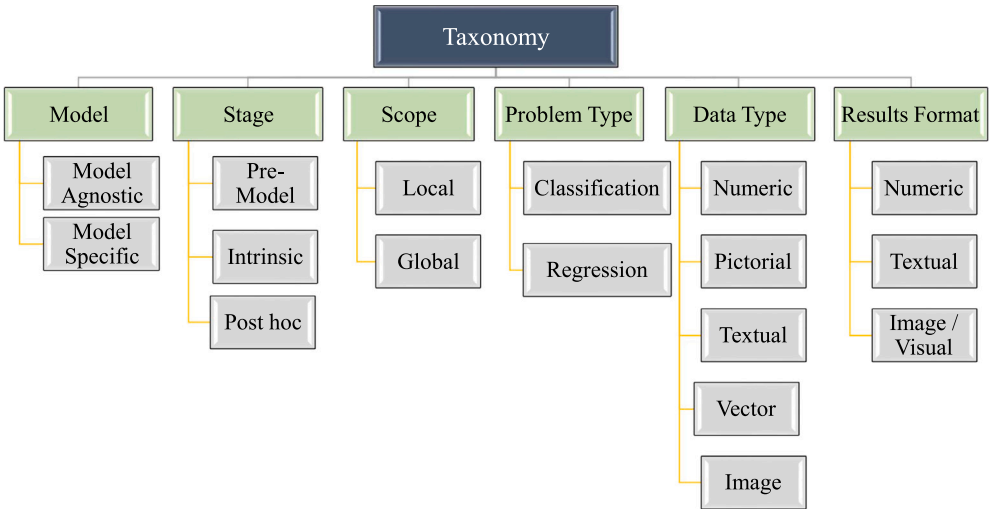


Fig. 1. Taxonomy of Explainable Artificial Intelligence (XAI).

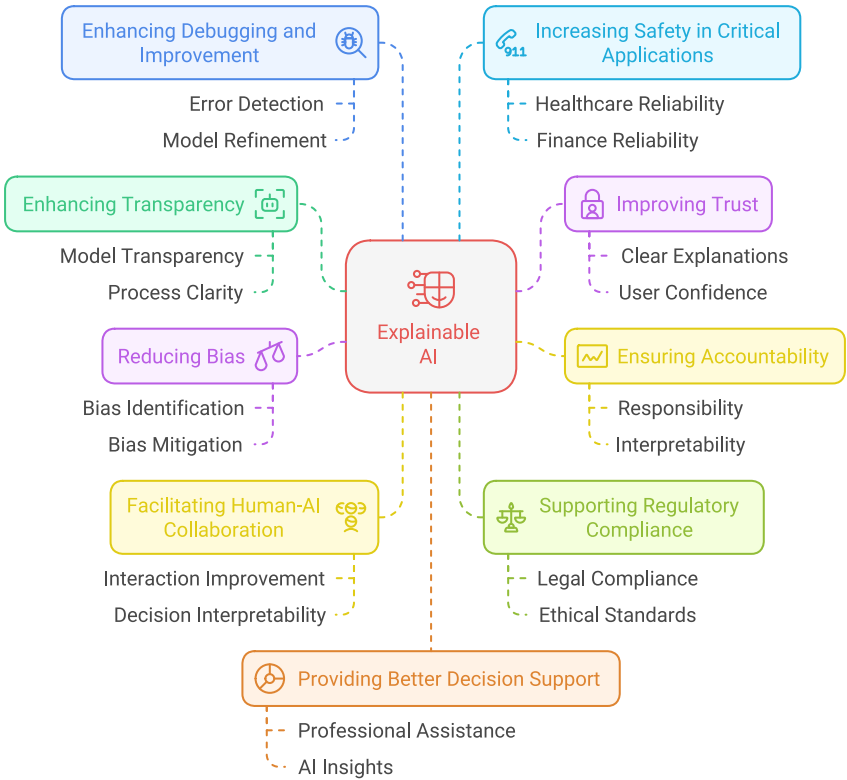


Fig. 2. Objectives and benefits of Explainable AI (XAI).

approaches and partially outlined their legal applications, financial, medical sectors, and transportation. Char et al. [46] evaluates the moral issues concerning the use of ML applications in healthcare (ML-HCAs). It provided a systematic approach to identifying these ethical concerns utilizing a flow model that incorporates the concept, development, and implementation stages of ML-HCA research as well as the related tasks of observation and assessment. Singh et al. [47] has produced a succinct description of explainable approaches for the applications of DL in image analysis in healthcare [48].

Existing SLRs sometimes overlook the nuances of medical imaging instead of covering broader XAI topics across multiple domains. Several aspects of XAI are frequently not integrated into a single SLR. As an example, whereas some SLRs may go into great detail on ML/DL methods, they might not sufficiently relate these algorithms to the particular

difficulties and datasets that are utilized in real-world applications, such as medical imaging [49]. This may lead to a lack of coverage of all pertinent elements required for a thorough grasp.

This SLR provides a full explanation of Grad-CAM's focus on research related to healthcare. It evaluates the standard performance metrics for evaluating XAI systems in medical applications and categorizes the most popular ML and DL algorithms for medical image processing. A detailed study of numerous improved Grad-CAM-related approaches was created specifically for more accurate XAI interpretation. These methods include Grad-CAM++, Guided Grad-CAM, Eigen-CAM, Mask Grad-CAM, and HayCAM. Table 1 compares this work to recent surveys by explaining and emphasizing major findings through synthesizing data utilizing an accurate mapping approach to demonstrate the research intensity in the selected topic area.

Table 1

Comparison of this Study with Existing Explainability Surveys. Symbols: X; Not utilized, ✓; discussed, ≈; discussed partially.

Ref.	Year	Focused applications	Machine learning	Deep learning	Datasets	Metrics	Grad-CAM in medical imaging	Grad-CAM and its variants	Frequency of publication
Holzinger et al. [50]	2017	Segmenting medical images and omic data	X	✓	X	X	X	X	X
Adadi and Berrada [45]	2018	Current developments in approaches that can be explained	✓	≈	X	X	X	X	X
Mathews [49]	2019	Uses of XAI in biology, NLP, and malware classification	✓	✓	X	X	X	X	X
Tjoa and Guan [43]	2020	Categorization of XAI methods and a synopsis of their medical applications	≈	≈	≈	X	X	X	X
Singh et al. [47]	2020	Using medical imaging to diagnose and forecast diseases	X	✓	≈	X	X	X	X
Char et al. [46]	2020	Finding moral dilemmas to use in the medical field	✓	≈	X	X	X	X	X
Adadi and Berrada [51]	2020	Creating interpretable models from black boxes	✓	≈	X	X	≈	X	X
Vilone and Longo [52]	2021	Explainability-related ideas and techniques for evaluating XAI	✓	≈	≈	✓	X	X	X
Rasheed et al. [53]	2022	ML for trustworthy and intelligible healthcare	✓	✓	≈	X	X	X	X
Borys et al. [54]	2023	XAI technique based on saliency for medical imaging professionals	X	✓	✓	X	X	X	✓
Haque et al. [55]	2023	XAI from a user's point of view	X	X	X	X	X	X	X
Allgaier et al. [56]	2023	A SLR on the explainability power of machine learning in healthcare	✓	✓	X	X	X	X	X
Hassan et al. [57]	2024	Model-Agnostic methods for medical image analysis using XAI	✓	✓	✓	✓	X	X	✓
Mohamed et al. [58]	2024	XAI for cancer diagnosis	✓	✓	✓	✓	X	X	X
Khan et al. [59]	2024	Moving toward trustworthy and reliable AI in healthcare	✓	✓	≈	≈	X	X	X
Bhati et al. [60]	2025	Analyzing various post-hoc explanation methods	✓	✓	≈	X	≈	X	X
This Study	2025	Model-Specific approach in XAI (Grad-CAM)	✓	✓	✓	✓	✓	✓	✓

2.3. XAI in medical imaging

Healthcare information is gathered from a variety of sources [61, 62]. While sensors collect certain data, clinical trials and research provide other data directly. Applications of AI and ML are crucial in the healthcare industry. Gaining the trust of AI algorithms is accomplished through XAI. It makes predictive models more reliable in practical situations. Identifying, preventing, and treating human illnesses are just a few of the many responsibilities that fall within the broad categories of healthcare and medicine [63]. The manual analysis of results in various areas, including computed tomography (CT) scans [64], ultrasounds [65], magnetic resonance imaging (MRI) [66], chest X-rays [67,68], and more, can occasionally leave clinicians feeling bored. The challenges of prevention and therapy are substantial for medical

professionals. While accurate identification is necessary for therapy, timely identification is necessary for prevention. ML models to aid in the process of diagnosis and are transparent and dependable, can help accomplish both of these objectives [69]. Sritharan et al. [70] introduced EnsembleCAM to make DL models more understandable when detecting cervical cancer from pap smear images. They used an XceptionNet model with 89% accuracy on the Herlev dataset. EnsembleCAM combines five popular explanation methods (GradCAM, GradCAM++, Score-CAM, Eigen-CAM, and LayerCAM) by taking the median of their results. This helped highlight the nucleus area more clearly, which is important for identifying cancer. They also used a pixel-flipping test to show that EnsembleCAM better identified the key regions the model used to make decisions.

The NeuroXAI framework was presented by Zeineldin et al. [71]. It is intended to examine brain pictures and incorporate explanation

approaches for showing maps that help with brain tumor diagnosis and detection in practical applications. Sepsis may arise from the body's reactivity to an infection, a potentially fatal medical condition. XAutoNet is an explainable, and extremely accurate AI model that was developed by Meena et al. [72] to use a few input characteristics to identify sepsis early. To enhance performance, the model employed an autoencoder based on CNN. In order to solve the black box problem of conventional AI models, the study integrates XAI for interpretability, providing visualization charts to medical staff to aid in diagnosing and therapy.

To increase local fidelity, Shankaranarayana and Runje [73] suggested an XAI method for breast cancer diagnosis called the autoencoder Local Interpretable Model-agnostic Explanations (LIME) technique. In order to provide a coherent explanation of the dataset, to make the forecast, the author employed a model of linear regression. Alkhalaf et al. [74] developed a technique that automated cancer detection in medical imaging using XAI technology to give patients and doctors a transparent and understandable decision-making process. The findings highlight the model's abilities as a reliable and effective technique for cancer detection and classification, highlighting its significance in enhancing the diagnostic process. Gamage et al. [75] proposed a DL system for classifying melanoma skin cancer using CNNs and Vision Transformers on the HAM10000 dataset. They used mask-guided segmentation with U2-Net and explained the model decisions using Grad-CAM and Grad-CAM++. The Xception-based CNN model achieved the highest accuracy of 98.37%. Shyamalee et al. [76] proposed a DL-based system to detect glaucoma using fundus images from the RIM-ONE dataset. They used an Attention U-Net with a ResNet50 backbone for segmenting the optic disc and cup, and a modified Inception V3 model for classifying glaucoma. The model achieved high accuracy in both segmentation and classification, with 98.97% accuracy for glaucoma prediction. To make the model more explainable, they used Grad-CAM and Grad-CAM++ to generate heatmaps showing which image areas influenced the diagnosis.

3. Gradient-Weighted Class Activation Mapping (Grad-CAM)

Grad-CAM is a model-specific technique for interpreting and visualizing DL models, particularly CNNs [21]. It enhances transparency by highlighting the most influential regions in an image that contribute to the model's decision. By leveraging gradients from the target class score relative to the last convolutional layer, Grad-CAM identifies key neurons that impact predictions. For instance, if a model classifies an image as a dog, Grad-CAM can reveal whether features like the head or paws played a decisive role. This approach provides valuable insights into the internal workings of deep CNNs, improving interpretability and trust in model decisions.

The rest of this section includes: Section 3.1 provides a mathematical formulation of Grad-CAM. Section 3.2 discusses key problems in XAI with Grad-CAM. Section 3.3 discusses the Application of Grad-CAM in medical imaging.

3.1. Grad-CAM mathematical formulation

The Grad-CAM is mathematically represented as:

$$L^c = \text{ReLU} \left(\sum_k q_k p_k \right) \quad (1)$$

where L^c is the heatmap for the target class c , p_k represents the activation map of the k th feature map from the target layer, and q_k is the weight associated with the k th feature map. The weight q_k is computed as:

$$q_k = \frac{1}{r_k} \sum_i \sum_j \frac{\partial L}{\partial p_k(i, j)} \quad (2)$$

where r_k denotes the number of elements in p_k , and $\frac{\partial L}{\partial p_k}$ represents the gradient of the class score L with respect to the feature map p_k . The function ReLU is the element-wise Rectified Linear Unit function, ensuring that negative values are set to zero.

This mathematical formulation encapsulates the fundamental processes of Grad-CAM, which entails a weighted aggregation of the activation maps, with the weights computed from the gradients, followed by the implementation of ReLU activation to emphasize the pertinent regions corresponding to the target class within the input image. Fig. 3 illustrates the internal mechanisms of Grad-CAM, wherein the gradients of each visual target within a classification network are examined in relation to the final convolutional layer, thereby generating a coarse localization map that delineates the significant areas of the image pertinent to concept prediction. This methodology can be employed within the context of the "Fully connected layer," "Structured output," and "Multimodal inputs" CNN model families, among others.

Grad-CAM generates an imprecise localization map that accentuates the significant regions of the image relevant to concept prediction through the utilization of the gradients associated with each target concept that propagate into the last convolutional layer. Grad-CAM affords the ability to visually validate the focal points of the network's attention, thereby confirming that it is indeed concentrating on the appropriate patterns within the image and activating in proximity to those patterns. The Class Activation Model (CAM) serves as a potent instrument for elucidating the operational dynamics of CNNs and fostering confidence in the developed applications; however, it is not devoid of certain constraints. A notable limitation of CAM is its requirement for feature maps to be positioned directly before the SoftMax layers, thereby restricting its applicability to a specific class of CNN architectures that execute global average pooling over convolutional maps immediately prior to making predictions.

3.2. Key problems in XAI with Grad-CAM

There are several problems in XAI using Grad-CAM [13,77]. While CNNs have outstanding performance in a variety of image-related tasks, comprehending how these models decide to act is hampered by their inherent complexity and opacity. The main issues in XAI using Grad-CAM for DL in image processing are illustrated in Fig. 4 and explained as follows:

- **Black-Box Models:** DL models, particularly deep neural networks, are frequently referred to as black-box models because of the complexity and difficulty of interpreting their internal workings. A basic difficulty in XAI using Grad-CAM for image processing is creating techniques to peer into these "black boxes" and derive relevant explanations [78].
- **Complexity and Non-linearity:** DL algorithms record extremely complex and non-linear correlations in image data, which makes it difficult to come up with clear, intelligible interpretations. A difficult task for Grad-CAM is creating methods that condense these intricate interactions into understandable forms without losing model performance [79].
- **High-Dimensional Data:** Since deep networks can have millions of parameters and images are high-dimensional data, it might be difficult for Grad-CAM to determine which attributes are important for making decisions. Creating meaningful explanations requires the development of techniques to decrease the dimensionality of visual data while maintaining essential information [80].
- **Performance-Interpretability Trade-off:** Increasing model performance frequently comes at the loss of interpretability. A model's efficacy may be harmed if it is simplified in search of better explanations. A persistent issue in XAI using Grad-CAM is finding the ideal balance between model performance and interoperability [81].

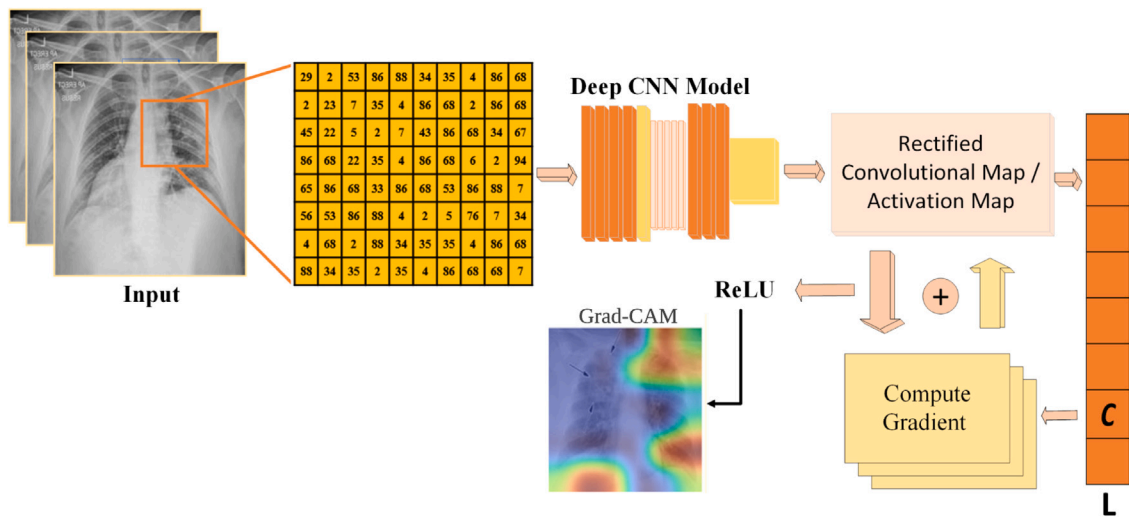


Fig. 3. The internal working of Grad-CAM.

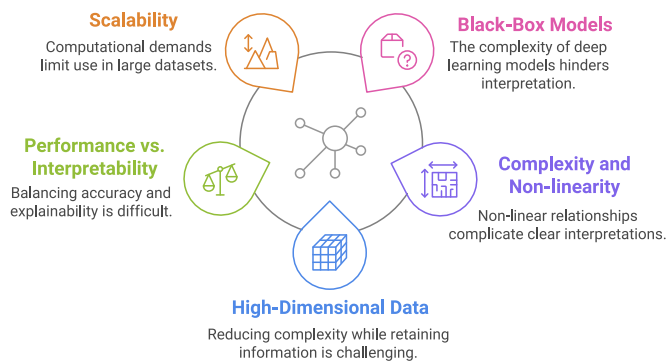


Fig. 4. Key Problems in XAI with Grad-CAM.

- **Scalability:** Scaling Grad-CAM to big datasets or real-time applications can be problematic since DL models for image processing can be computationally demanding [82]. Practically speaking, it is difficult to find effective XAI techniques with large-scale image data. Grad-CAM is not utilized for 2D CNNs [83]. It has also been efficiently applied to 1D CNNs, showing an understanding of significant factors that contribute [24].

3.3. Grad-CAM in medical imaging

Explainable DL has emerged as a pivotal domain of inquiry within the field of AI, particularly in the realm of medical imaging, where the necessity for interpretability is paramount for effective diagnosis and treatment planning. Grad-CAM, a prevalent methodology employed for visualizing the decision-making processes inherent in DL models, has substantiated its significance in augmenting model transparency. Recent investigations have assessed its efficacy across a spectrum of medical imaging tasks, elucidating its contribution to enhancing model interpretability and fostering trustworthiness [84]. A noteworthy obstacle in the application of medical AI lies in the deficiency of explainability, which impedes its integration into clinical practice. While methodologies based on DL have exhibited remarkable efficacy in medical imaging, the relational nature of these models imposes constraints on their transparency. Grad-CAM, which produces heatmaps to delineate class-specific activation regions, has emerged as a promising strategy to mitigate this concern. Research endeavors have explored

whether AI-generated heatmaps can assist radiologists in the diagnostic process, with findings indicating that Grad-CAM visualizations are generally advantageous for medical practitioners [85]. Implementing XDL methodologies extends to vital medical challenges such as diagnosing oral cancer, wherein early detection is crucial. Although DL models have showcased elevated classification accuracy, their inherent lack of interpretability necessitates supplementary techniques such as Grad-CAM for visual elucidations. Investigations that integrate subclass accuracy assessments and prediction confidence distributions have facilitated a more holistic comprehension of model performance, particularly within the context of oral cancer detection [86].

Notwithstanding its merits, Grad-CAM and analogous saliency methodologies possess limitations that necessitate resolution before clinical implementation. A comparative examination of various saliency techniques has indicated that while Grad-CAM excels in pathology localization, it still falls short in comparison to expert radiologists. The disparity in performance is particularly pronounced for smaller and intricately shaped pathologies, underscoring the imperative for additional advancements in saliency-based explainability techniques [87]. Moreover, initiatives aimed at augmenting explainability within DNNs have culminated in the formulation of sophisticated techniques such as Grad-CAM++. Research concentrating on the detection of shoulder abnormalities has illustrated that Grad-CAM++ can enhance visual interpretations by refining activation maps. By integrating gradient-based modifications, this methodology bolsters transparency, ultimately nurturing greater confidence in AI-driven medical diagnostic systems [88]. Collectively, these investigations emphasize the critical importance of elucidating AI within the domain of medical imaging and the continuous endeavors to enhance methodologies such as Grad-CAM. Although these approaches facilitate enhanced interpretability, obstacles persist in reconciling the disparity between AI-driven decision-making and the assessments rendered by human experts, thereby necessitating additional progress in the development of explainability frameworks. Dasanayaka et al. [89] developed a tool called Tumor-Analyzer that combines MRI and whole slide images to classify brain tumors while ensuring interpretability through Grad-CAM and attention mechanisms. Their approach enhances transparency and allows clinicians to visually understand the model's decision-making process, reducing the black-box nature of DL in healthcare.

4. Methodology

The current SLR was performed using the standards set by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses



Fig. 5. Studies mapping process.

(PRISMA) [90]. The study also adhered to the standard recommendations proposed by Keele et al. [91]. The methodology conceptual framework used in this research is based on SLRs proposed according to Al-Selwi et al. [92], Talpur et al. [93], Sumiea et al. [94].

The mapping procedure employed in this SLR comprises of seven connected stages, as indicated in Fig. 5 and further discussed in Baseline Study (4.1), Research Question Alignment (4.2), Search Criteria (4.3), Publication Screening (4.4), Quality Assessment of Selected Studies (4.5), Data Extraction (4.6), and Compilation of Included Studies (4.7).

4.1. Baseline study

A Baseline study is the first and most important step in a comprehensive literature review. At this stage, we focus on limiting the search criteria by gathering background knowledge about Grad-CAM research. A subject search was conducted on vital search engines around the World Wide Web using the same specific phrase, such as Grad-CAM AND XAI. After that, we reduced search outcomes and focused on a study that supported us in identifying keywords specifically related to the topic. Keyword analysis provides an overview of a study by briefly defining the content and previous research patterns in a certain subject. Co-occurrence analysis on author keywords was carried out in this bibliometric review of XAI and Grad-CAM in Medical Imaging using VOSviewer [95], a bibliometric analysis software tool. Keyword co-occurrence mapping and networks discover emerging ideas in a knowledge domain over time by assessing the domain's research development. The 427 chosen publications' research information system (RIS) format data file was imported into VOSviewer for keyword co-occurrence analysis. The minimal keyword occurrence was set to 10; therefore, out of the 413 keywords that were obtained, 28 met the criterion. For better representation, these broad keywords — training, classification model, data, computer vision, XAI, and Grad-CAM method were used. The co-occurrences of keywords are mapped into numerous clusters in Fig. 6, each of which depicts the network connecting related terms. The mean year of the publications a keyword occurs in is its average publication year in VOSviewer.

4.2. Research question alignment

Primary research questions are defined to narrow the study's focus and writing process. The evaluation is based on the subject that is related to the focus area and specific concern. As a result, significant research has been performed into the creation of XAI techniques. However, it is difficult to obtain enough literature information on new Grad-CAMs that use DL approaches such as AlexNet, ResNet, and

Darknet. As a result, the fundamental goal of this work is to conduct an in-depth assessment of Grad-CAM development, optimization techniques, and application subjects. The subsequent are the primary research questions:

1. **RQ1:** What are the different variations of Grad-CAM?
Motivation: Exploring the variations of Grad-CAM will reveal enhancements and modifications that improve its performance and usability in various contexts.
2. **RQ2:** What are the most popular ML/DL algorithms that researchers employ with Grad-CAM?
Motivation: Identifying the commonly used algorithms will provide insight into the compatibility and applicability of Grad-CAM with various ML/DL methods.
3. **RQ3:** What are the most often used performance metrics in the ML/DL methods utilizing Grad-CAM?
Motivation: Understanding the performance metrics will help in evaluating the effectiveness and accuracy of ML/DL models in different applications.
4. **RQ4:** What is the publication frequency in the field of the Grad-CAM model?
Motivation: The issue aims to determine the frequency of publications and current research trends related to Grad-CAM by allocating the included studies according to the publishers, the publication type, and the yearly trend.

4.3. Search criteria

The initial search was performed using Scopus, Web of Science, and Science Direct search engines, which assisted in the development of the research topics for this study. To find appropriate keywords and search strings associated with the Grad-CAM problem, the study used research questions. The Baseline Study stage provided us the opportunity to use the criteria for searching to look at the problem from a broader viewpoint. In SLR analysis, four databases and engines were selected for a search within the Grad-CAM sector utilizing completed keywords. The databases have been selected to provide a wide range of the most significant and high-impact journals and conference proceedings. Although each research database uses its search features and constraints to conduct a systematic search, it is necessary to personalize the keyword search to each database. Table 2 and Fig. 7 displays the search parameters utilized to perform a systematic search for the study using the given databases.

The research was conducted to identify relevant studies distributed across multiple databases using predefined keywords combined with the logical operator AND. It was important to adapt the search terms to the specific indexing features of each database. The research questions were systematically formulated to analyze the productivity and application of Grad-CAM in scholarly publications from 2020 to 2024. The final search strategy employed the keywords "Grad-CAM" AND "XAI", and the last search was completed on December 26, 2024, ensuring comprehensive coverage within the defined timeframe. Preprint platforms (e.g., arXiv, bioRxiv) were excluded to ensure that only peer-reviewed and validated studies were considered.

4.4. Publication screening

All publications that had been found after the database search was completed with the selected keywords were put through an extensive screening procedure based on inclusion as well as exclusion. This process was designed to filter and select only those studies that were important to addressing the research questions, while systematically removing studies that were judged irrelevant. Fig. 8 outlines the criteria used to evaluate each article critically and determine whether it should be included or excluded in this SLR. Table 3 refers to the thorough process of methodical article screening carried out in four different databases.

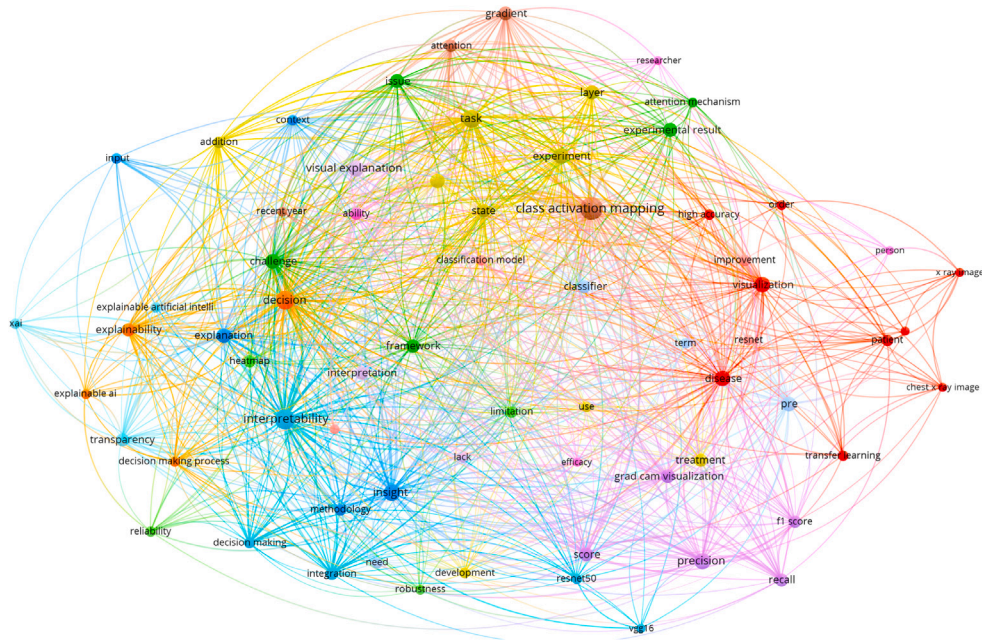


Fig. 6. Co-occurrence mapping of the most used keyword.

Table 2
SLR search measures.

No.	Databases	Systematic search framework	Keywords arrangement	Category of publication	Time frame
1	Scopus	Title, abstract, and keywords	“Grad-CAM” AND “XAI”	Journals Papers & Conferences.	2020 to 2024
2	Web of Science				
3	Science Direct				
4	IEEE Xplore				

Table 3
Systematic screening of articles.

Stages	Web of Science	Scopus	ScienceDirect	IEEE Xplore	Total
Collected Studies	163	150	36	78	427
Filtered Results	80	44	21	27	172
Duplicate Papers	32 Excluded				140
	Screening				
Relevant to Our Focus	89 Excluded				51
	Included				

4.5. Quality assessment of selected studies

In accordance with the defined inclusion and exclusion criteria, it is critical to assess the eligibility and relevance of the primary studies selected in the previous stage. In this investigation, three distinct quality assessment scores utilized to address each inquiry were derived from Al-Selwi et al. [92] to scrutinize the eligibility of individual studies based on various criteria, including the significance of a specific study, the caliber of the results and analyses, and recommendations or findings relevant to future research. The scoring method consists of 1 (Yes), 0.5 (Partially), and 0 (No), with quality assessment ratings used to evaluate the acceptability of every manuscript in the gathered data. Fig. 9 shows the quality score distribution of the publications, in which the blue columns indicate the included studies.

4.6. Data extraction

In Stage 4.4 and Stage 4.5, initial publication screening and Quality Assessment of selected studies were performed. The selection process

of included studies is also discussed in Stage 4.4 and Stage 4.5. The processes of data extraction and compilation are used to analyze the selected studies. Usually, data extraction and synthesis methods are carried out using a variety of accessible tools and applications, including Google Sheets, REDCap, and Microsoft Excel. Microsoft Excel is used to store data from publications. The following information was carefully extracted from every single study included in the analysis: title, abstract, keywords, authorship, year of publication, scientific databases, type of publication, such as journal article, conference proceeding, or book chapter, and study categorization. Fig. 10 displays the flow of information across several phases as specified in the SLR, using the PRISMA framework proposed by Haddaway et al. [96].

4.7. Compilation of included studies

In this last stage, all studies related to our focus are included. First, all duplicate studies are excluded. A Microsoft Excel sheet was created to main all studies from different scholarly databases. These studies involve details like the name of the journal, title, author name, venue,

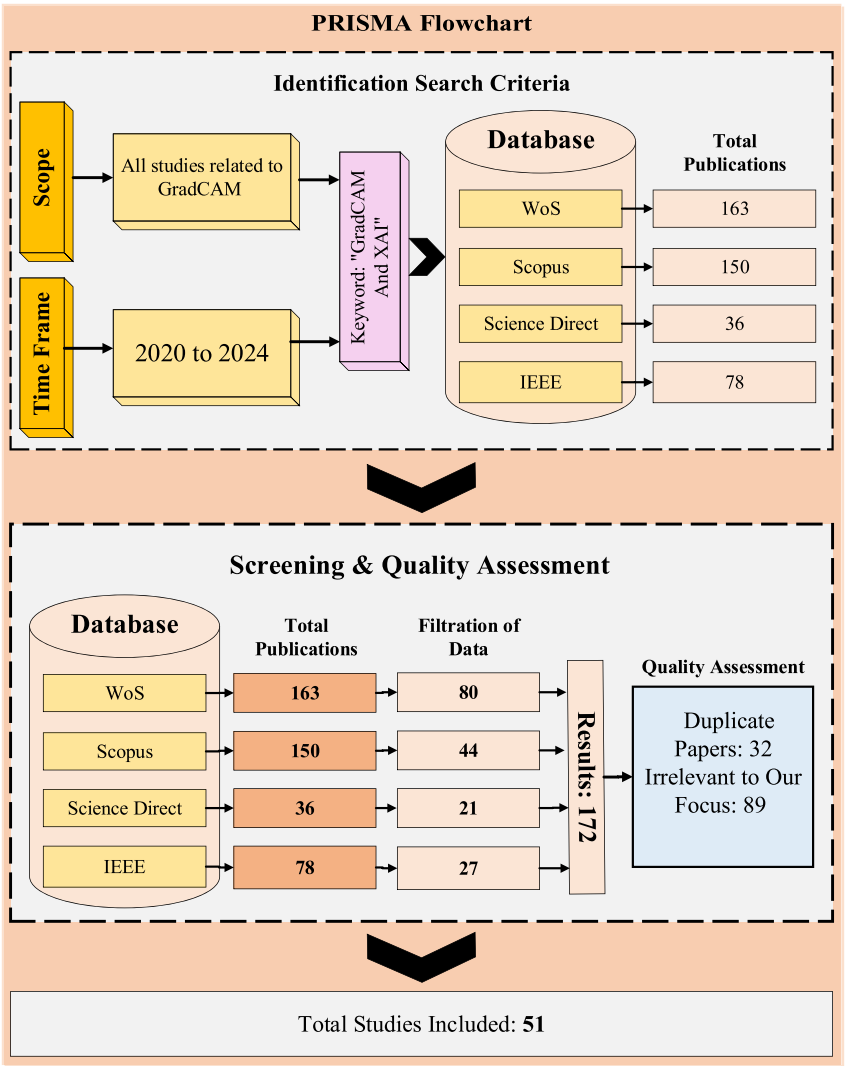


Fig. 7. The PRISMA block diagram of the SLR.

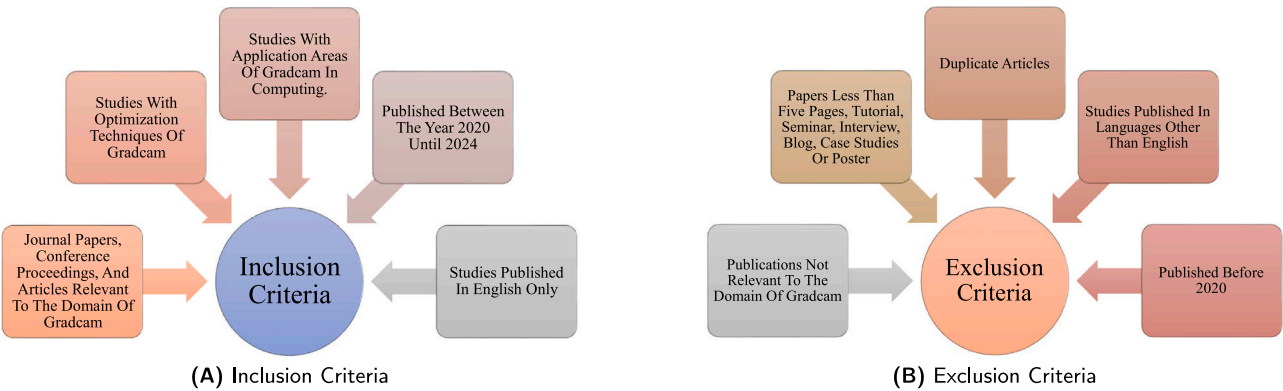


Fig. 8. Search Screening: (A) Inclusion Criteria, and (B) Exclusion Criteria.

DOI, paper type (such as review, conference, book chapter, or journal), and year of publication. The included studies and preferences are given to the article related to Grad-CAM, published between 2020 and 2024, written in the English language. Furthermore, after this extensive mapping process, the last 51 studies were collected and included in this SLR to satisfy all the mentioned research questions. These studies

help us identify research gaps, challenges, and concerns in the field of Grad-CAM.

Fig. 11 illustrates the distribution of academic databases used in this SLR. It shows that the majority of articles (39%) were sourced from Scopus, followed by 25% from Web of Science, 20% from ScienceDirect, and 16% from IEEE Xplore. This indicates a diverse selection

Algorithm 1: Grad-CAM (Gradient-weighted Class Activation Mapping)**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.

Result:

- $L_{\text{Grad-CAM}}^c$: Class activation map for the target class c .

- 1 **Forward Pass:** Pass the input image I through the CNN model to obtain feature maps A^k at the last convolutional layer.
- 2 **Compute Class Score:** Compute the score for the target class c , denoted by y^c , before the softmax layer. $y^c \leftarrow \text{score}(c)$
- 3 **Compute Gradients:** Compute the gradients of the score y^c with respect to the feature maps A^k .
- 4 $\frac{\partial y^c}{\partial A^k} \leftarrow$ Compute gradient
- 5 **Global Average Pooling:** Perform global average pooling on the gradients to obtain weights α_k^c .
- 6 $\alpha_k^c \leftarrow \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$ // Where Z is the number of pixels in A^k
- 7 **Compute Class Activation Map:** Compute the weighted sum of the feature maps using the weights α_k^c . Apply ReLU to the result.
- 8 $L_{\text{Grad-CAM}}^c \leftarrow \text{ReLU}(\sum_k \alpha_k^c A^k)$
- 9 **Upsample (Optional):** Optionally, upsample $L_{\text{Grad-CAM}}^c$ to match the size of the input image I .
- 10 **Return:** The class activation map $L_{\text{Grad-CAM}}^c$.



Fig. 9. Quality score distribution of the publications.

of databases, ensuring comprehensive coverage of relevant literature across different platforms.

5. Comparative analysis and data synthesis

In this section, the research questions listed in Section 4.2 are discussed and answered. It depicts evidence related to the current studies and their challenges. It presents all related studies associated with Grad-CAM.

5.1. RQ1: What are the different variations of Grad-CAM?

Grad-CAM [21,97] is a significant technique in the field of XAI, particularly in visualizing the areas of input images that are important for the predictions of CNNs as described in Algorithm 1. This technique aids in understanding and interpreting the decision-making process of DL models, thereby enhancing transparency and trust in AI systems. In examining the various adaptations of Grad-CAM as discussed in Table 4, a range of variations have been developed to improve its effectiveness and applicability.

From the extracted data of the included papers in this SLR, Fig. 12 shows the most widely used variations of Grad-CAM. While the original Grad-CAM remains the dominant method, several other variations are also utilized. The original Grad-CAM is employed in 67% of the instances ($n=38$). This is followed by Grad-CAM++, which represents 16% ($n=9$) of the instances, indicating its significant yet secondary usage. Guided Grad-CAM is another notable variant, accounting for 5%

($n=3$) of the cases. Several other variations are also utilized, albeit less frequently. These include Ablation-CAM, CAM, EigenGrad-CAM, Grad-CAM-based RISE (GRIPS), HaGrad-CAM, Layer-CAM, Mask Grad-CAM, and Random-CAM, each constituting 2% or less of the total, with only one instance ($n=1$) each.

5.1.1. CAM

Class Activation Mapping (CAM) is a foundational technique for visualizing the regions of an image that a CNN uses to make a decision. In the CAM approach, a global average pooling layer is applied just before the final classification layer of a CNN. This pooling layer ensures that each feature map corresponds to a particular class. A class activation map is generated by weighing these feature maps according to the weights of the classification layer and summing them up. This map highlights the discriminative regions of the image that contribute the most to the class prediction. The key contribution of CAM is its ability to produce interpretable visual explanations of CNN decisions without requiring any modifications to the architecture or additional training. This technique paved the way for more advanced methods like Grad-CAM and its variations, which further enhance the interpretability and applicability of DL models in various domains. By understanding the importance of CAM and its original implementation, researchers and practitioners can better appreciate the evolution of XAI techniques and their impact on making AI systems more transparent and trustworthy. The algorithm based on the Class Activation Mapping (CAM) is explained in Algorithm 2.

5.1.2. Grad-CAM++

Grad-CAM++ by Chattopadhyay et al. [148] is an enhancement of the original Grad-CAM technique, designed to address some of the limitations of its predecessor. The original Grad-CAM, as introduced by Selvaraju et al. (2017), helps visualize the regions of an input image that a CNN focuses on when making a prediction. While powerful, Grad-CAM has some drawbacks, particularly in accurately attributing the importance of individual pixels in complex images or in cases of overlapping objects. One of the main issues with the original Grad-CAM is its tendency to produce coarse and less precise localization maps, especially when dealing with multiple objects or fine-grained details in an image. This is caused by Grad-CAM combines the gradients of the target class in relation to the feature maps, which leads to less clean representations. Chattopadhyay et al. (2018) created Grad-CAM++ to address these limitations. Grad-CAM++ calculates pixel-wise weights for each neuron in the feature maps. This method allows for a more detailed and fine-grained attribution of pixel importance, resulting in sharper and more precise localization maps. This helps in distinguishing

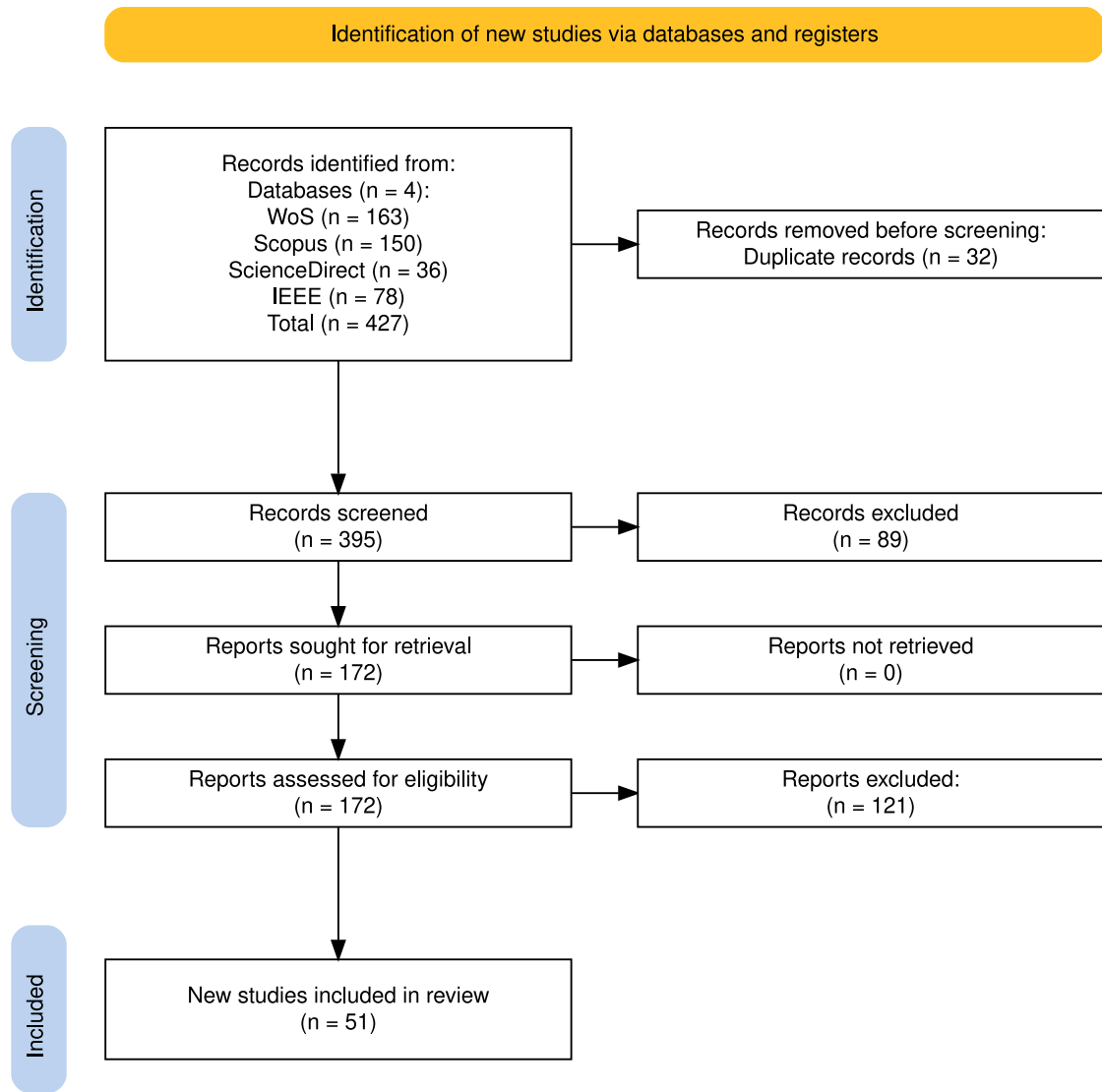


Fig. 10. The PRISMA flowchart of the SLR.

Algorithm 2: CAM (Class Activation Mapping)**Data:**

- I : Input image.
- CNN: CNN model with a global average pooling layer before the fully connected layer.
- c : Target class.

Result:

- L_{CAM}^c : Class activation map for the target class c .

- 1 **Forward Pass:** Pass the input image I through the CNN model to obtain feature maps A^k at the last convolutional layer.
- 2 **Global Average Pooling:** Apply global average pooling to the feature maps A^k to obtain the pooled features.
- 3 $F^k \leftarrow \frac{1}{Z} \sum_i \sum_j A_{ij}^k$ // Where Z is the number of pixels in A^k
- 4 **Fully Connected Layer:** Compute the score for the target class c using the weights w_k^c of the fully connected layer.
- 5 $y^c \leftarrow \sum_k w_k^c F^k$
- 6 **Compute CAM:** Generate the class activation map by computing the weighted sum of the feature maps using the weights w_k^c .
- 7 $L_{CAM}^c \leftarrow \sum_k w_k^c A^k$
- 8 **Upsample (Optional):** Optionally, upsample L_{CAM}^c to match the size of the input image I .
- 9 **Return:** The class activation map L_{CAM}^c .

between closely situated objects and in highlighting finer details within an image. Grad-CAM++ is more effective in scenarios where multiple instances of an object are present in an image. By using a weighted

combination of the gradients, Grad-CAM++ can better capture the contribution of each pixel to the class score, even when multiple objects overlap or are in close proximity. The enhanced weighting mechanism

Table 4

Related studies of XAI interpretation method using grad-CAM and its variations.

Ref.	Dataset	ML/DL techniques	XAI techniques	Evaluation metrics
Mettivier et al. [98]	Tomosynthesis datasets	DCNN	Grad-CAM	AUC, Accuracy, Precision, Sensitivity, Specificity, F1-Score, False Positive Rate
Selvaraju et al. [99]	Dataset bias	CNN, AlexNet, VGG, ResNet	Grad-CAM	Accuracy
Schöttl [100]	PASCAL VOC 2012 dataset	ResNet-50	Grad-CAM	Accuracy
Ennadifi et al. [101]	Proposed dataset	CNN, Mask R CNN, DenseNet-121	Grad-CAM	Accuracy, Training time
Kusunose et al. [102]	N/A	CNN, ResNet-32	Grad-CAM-based RFS (GRFS)	Accuracy, Precision, Recall
Prasad et al. [103]	MIT-BIH ARRHYTHMIA DATASET	CNN, EfficientNet-B3	Grad-CAM	Specificity, Sensitivity, F1-Score, AUC
Prasad et al. [103]	OASIS-3 dataset	HighRes3DNet	Grad-CAM	MSE Loss
Cai et al. [104]	DCASE2022task4 dataset	CNN	Grad-CAM	PSDS2
Maximino et al. [105]	POCOVID	VGG16, ResNet-50, DenseNet-169, MobileNet-v2	Grad-CAM	Accuracy
Nasiri et al. [106]	ChestX-ray8 dataset	LightGBM, DenseNet-169, MobileNet	Grad-CAM	Sensitivity, Specificity, Precision, F1-Score, Accuracy
El Mohamadi and El Hassouni [107]	CXR images	CNN, VGG16, ResNet-50	FEM, Grad-CAM	Precision, Recall, F-Score, Accuracy, Training Loss
Rifat et al. [108]	Knee osteoarthritis severity grading dataset	DenseNet-169, Inception-v2, MobileNet-v2	Grad-CAM	Accuracy, Precision, Recall, F1-Score
Golash et al. [109]	PlantVillage dataset	ResNet-101, EfficientNet-B7, Vision Transformer	Guided Grad-CAM	Accuracy, Training Loss
Tan et al. [110]	ImageNet2012 dataset	VGG16	Grad-CAM, NoiseCAM	Accuracy
Rajaraman and Shanmugam [111]	X-ray MRI and CT scans	CNN	Grad-CAM++	Accuracy, Precision, Recall, F1-Score
Jitha et al. [112]	Pneumonia dataset	CNN	Grad-CAM	Accuracy, Training Loss
Rashmi et al. [113]	ALZHEIMER'S DATASET	DINO-v2, ResNet-101, DenseNet-121, VGG16	Grad-CAM	Accuracy, F1-Score, Precision, Recall
Bhat and Raychowdhury [114]	CIFAR-10 dataset	CNN, ResNet-18	Grad-CAM	Accuracy, Latency (ms)
Collazos-Huertas et al. [115]	Giga-Science	CNN	Grad-CAM++	Accuracy
Chowdhury et al. [116]	COVID CXR Image Dataset (Research)	EfficientNet, DenseNet-121, VGG16, VGG19, Inception-v3	Grad-CAM	Accuracy, Precision, Recall, F1-Score
Bajhaiya and Unni [117]	kvasir-capsule dataset	CNN, GINet	Grad-CAM, Guided Grad-CAM	Sensitivity, Specificity
Jamil et al. [118]	EyePacs dataset	CNN, VGG16	Grad-CAM++	IoU
Liu et al. [119]	QGIS	GLDCNet	Grad-CAM	Recall, F1-Score, Accuracy, Precision
Hroub et al. [120]	Chest X-ray datasets	ResNet-50, Inception-v3, DenseNet-201, EfficientNet-B7, MobileNet-v3	Grad-CAM, Grad-CAM++, EigenGrad-CAM, Ablation-CAM, Random-CAM	N/A
Jain et al. [121]	KID dataset	CNN, WCENet	Grad-CAM++	Accuracy, Precision, Recall, F1-Score, AUC
Schneider and Vlachos [122]	Fashion-MNIST	CNN, ResNet, VGG	Grad-CAM	Accuracy
Taşcı [123]	Brain tumor dataset	DenseNet-201	DGXAINet, Grad-CAM	Accuracy
Karki et al. [124]	CXR dataset	CNN	Grad-CAM	AUC, Accuracy
Tummala et al. [125]	LC25000 histopathology dataset	EfficientNet-v2	Grad-CAM, saliency map	Accuracy, AUC, F1-Score, BA, MCC
Kumar et al. [126]	Malaria dataset	CNN, Mosquito Net	Grad-CAM	Accuracy, AUC, Sensitivity, Specificity, F1-Score, MCC
Jain et al. [127]	Dementia dataset	CNN, VGG16, VGG19	Grad-CAM	Precision, Recall, F1-Score, Accuracy

(continued on next page)

Table 4 (continued).

Ref.	Dataset	ML/DL techniques	XAI techniques	Evaluation metrics
Varam et al. [128]	Kvasir-Capsule Dataset	Inception-v3, EfficientNet-v2, VGG16, VGG19, MobileNet-v3, ResNet-152v2	Grad-CAM, Grad-CAM++, LayersCAM, LIME, SHAP	N/A
Ashwath et al. [129]	PAD-UFES-20 dataset	DenseNet-121, Inception, Xception, ResNet-50	TS-CNN	Accuracy, Precision, Recall, F1-Score, AUC
Lai et al. [130]	Whole Slide Images (WSIs) dataset	CNN	Grad-CAM	Accuracy
Nguyen et al. [131]	ILSVRC-2012	ResNet-50, Inception-v3, VGG16	SeCAM	Accuracy
Turan et al. [132]	Pavia University dataset	CNN	Grad-CAM, Grad-CAM++, Guided Backpropagation	Visual representation
Singh et al. [133]	Ultrasound images	VGG16, ResNet-50, GoogLeNet, Xception, SqueezeNET	Grad-CAM	Accuracy, Sensitivity, Specificity, F1-Score, Cohen's kappa, AUC
Rani et al. [134]	COVID-Pneumonia CXR	EXP-Net	Grad-CAM++	AUC, Cohen's kappa, F1-Score, Training Loss, Accuracy, Precision, Recall
Park and Wallraven [135]	AffectNet dataset	DNN	CAM, Grad-CAM	Accuracy
El Habib Daho et al. [136]	SARS-CoV-2 CT-scan	ResNet-50, DenseNet-121	Grad-CAM	Accuracy, Sensitivity, Specificity, F1-Score
Punn and Agarwal [137]	COVID-19 CT	CHS-Net	Grad-CAM++	Accuracy, Precision, Specificity, Recall, Dice coefficient, Jaccard
Chang et al. [138]	Glaucoma	DLMs	Grad-CAM	AUC, Sensitivity, Specificity
Bhatt et al. [139]	SIPaKMeD	VGG19, ResNet-34, ResNet-101, EfficientNet-B3, EfficientNet-B4, EfficientNet-B5	Grad-CAM	Accuracy, Precision, Recall, F-Beta Score, Cohen's kappa
Schöttl [140]	Imagenet ILSVRC 2012	ResNet-50	Grad-CAM	Accuracy
Niranjan et al. [141]	CT scan dataset	CNN, Res2Net	Guided Grad-CAM	Accuracy, Precision, Recall, F1-Score
Inbaraj and Jeng [142]	N/A	Mask R CNN	Mask Grad-CAM	Accuracy
Palatnik de Sousa et al. [143]	COVID-CT dataset	VGG16, DenseNet, EfficientNet	Grad-CAM, LIME, RISE, Squaregrid	Accuracy, F1-Score, AUC
De Silva et al. [144]	Asian elephants image dataset	You Only Look Once (YOLO), CNN, VGG16, ResNet-50, Inception-v3, AlexNet	Grad-CAM	Accuracy
Sharmila et al. [145]	Corn Leaf Disease and Skin Cancer Disease datasets	CNN, ResNet, MobileNet	Grad-CAM	Accuracy, Memory size
Örnek and Ceylan [146]	Face Mask detection Dataset	CNN	HayCAM	Accuracy
Rahman et al. [147]	Ultrasound images	PReLU-Net, SqueezeNET, Swin Transformer	LIME, Grad-CAM	Accuracy, F1-Scores, Precision, Recall

in Grad-CAM++ provides better visual explanations by more accurately highlighting the regions that influence the CNN's predictions. This leads to increased interpretability and trust in the model's decision-making process. The algorithm based on the Grad-CAM++ is explained in Algorithm 3.

5.1.3. Guided Grad-CAM

Guided Grad-CAM by Selvaraju et al. [99] is an extension of the Grad-CAM technique, developed to provide more precise and interpretable visual explanations of CNN predictions. However, Grad-CAM alone may produce somewhat coarse and low-resolution visualizations due to the inherent spatial down-sampling in CNNs, making it less effective for detailed interpretation. Guided Grad-CAM, as described by Selvaraju et al. addresses this issue by combining the class-discriminative localization map of Grad-CAM with guided backpropagation, which is a technique that provides fine-grained visualizations by showing the gradients of the output neuron with respect to the input image, but only passing positive gradients. This combination results in high-resolution and detailed visual explanations that are both class-discriminative and

pixel-precise. The primary problem addressed by Guided Grad-CAM is the trade-off between spatial resolution and class discrimination. Traditional Grad-CAM can effectively highlight the important regions corresponding to a particular class, but it lacks fine detail. On the other hand, guided backpropagation offers detailed gradients but is not class-specific. Guided Grad-CAM integrates the strengths of both methods, providing detailed and class-specific visualizations. This hybrid approach resolves the issue of coarse localization by overlaying the fine-grained details from guided backpropagation on the class-specific heatmaps from Grad-CAM, resulting in a more comprehensive and interpretable visualization. The algorithm based on the Guided Grad-CAM is explained in Algorithm 4.

5.1.4. Eigen-CAM

Eigen-CAM [149,150] is an innovative technique aimed at enhancing the interpretability of CNNs. Eigen-CAM addresses the limitations of traditional methods like Grad-CAM, which often produce noisy and less interpretable activation maps, especially in complex images. These noisy maps can obscure the important regions that influence the

Algorithm 3: Grad-CAM++ (Gradient-weighted Class Activation Mapping Plus Plus)**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.

Result:

- $L_{\text{Grad-CAM++}}^c$: Class activation map for the target class c .

- 1 **Forward Pass:** Pass the input image I through the CNN model to obtain feature maps A^k at the last convolutional layer.
- 2 **Compute Class Score:** Compute the score for the target class c , denoted by y^c , before the softmax layer.
- 3 $y^c \leftarrow \text{score}(c)$
- 4 **Compute First-Order Gradients:** Compute the first-order gradients of the score y^c with respect to the feature maps A^k .
- 5 $\frac{\partial y^c}{\partial A^k} \leftarrow$ Compute first-order gradient
- 6 **Compute Second-Order Gradients:** Compute the second-order gradients of the score y^c with respect to the feature maps A^k .
- 7 $\frac{\partial^2 y^c}{\partial (A^k)^2} \leftarrow$ Compute second-order gradient
- 8 **Compute Third-Order Gradients:** Compute the third-order gradients of the score y^c with respect to the feature maps A^k .
- 9 $\frac{\partial^3 y^c}{\partial (A^k)^3} \leftarrow$ Compute third-order gradient
- 10 **Compute Alpha Weights:** Compute the pixel-wise importance weights α_{ij}^k for each feature map A^k as follows:
- 11

$$\alpha_{ij}^k \leftarrow \frac{\partial^2 y^c}{\partial (A_{ij}^k)^2} \bigg/ 2 \cdot \frac{\partial^2 y^c}{\partial (A_{ij}^k)^2} + \sum_p \sum_q A_{pq}^k \cdot \frac{\partial^3 y^c}{\partial (A_{pq}^k)^3}$$

// Where i, j are pixel indices, and p, q sum over all pixels in the feature map.

- 12 **Compute Weights:** Compute the final weights α_k^c for each feature map A^k by averaging α_{ij}^k over all pixels i, j .
- 13

$$\alpha_k^c \leftarrow \sum_i \sum_j \alpha_{ij}^k \cdot \text{ReLU} \left(\frac{\partial y^c}{\partial A_{ij}^k} \right)$$

- 14 **Compute Class Activation Map:** Compute the weighted sum of the feature maps using the weights α_k^c . Apply ReLU to the result.
- 15 $L_{\text{Grad-CAM++}}^c \leftarrow \text{ReLU} \left(\sum_k \alpha_k^c A^k \right)$
- 16 **Upsample (Optional):** Optionally, upsample $L_{\text{Grad-CAM++}}^c$ to match the size of the input image I .
- 17 **Return:** The class activation map $L_{\text{Grad-CAM++}}^c$.

Algorithm 4: Guided Grad-CAM (Guided Backpropagation combined with Grad-CAM)**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.

Result:

- $L_{\text{Guided Grad-CAM}}^c$: Guided Grad-CAM visualization for the target class c .

Step 1: Grad-CAM Calculation

- Perform forward pass: Pass the input image I through the CNN model to obtain feature maps A^k at the last convolutional layer.
- Compute class score: Calculate the score for the target class c , denoted by y^c , before the softmax layer.
- Compute gradients: Calculate the gradients of the score y^c with respect to the feature maps A^k .
- Global average pooling: Perform global average pooling on the gradients to obtain weights α_k^c .
- Compute class activation map: Compute the weighted sum of the feature maps using the weights α_k^c . Apply ReLU to the result to obtain the Grad-CAM heatmap $L_{\text{Grad-CAM}}^c$.

Step 2: Guided Backpropagation

- Modify backpropagation: Modify the backpropagation algorithm to pass only positive gradients through the ReLU layers.
- Compute guided gradients: Perform a backward pass from the target class c to the input image I , using the modified backpropagation to obtain the guided gradients G_{Guided}^c .

Step 3: Combine Grad-CAM and Guided Backpropagation

- Element-wise multiplication: Perform an element-wise multiplication of the Grad-CAM heatmap $L_{\text{Grad-CAM}}^c$ and the guided gradients G_{Guided}^c to obtain the final Guided Grad-CAM visualization $L_{\text{Guided Grad-CAM}}^c$.

Return: The final Guided Grad-CAM visualization $L_{\text{Guided Grad-CAM}}^c$.

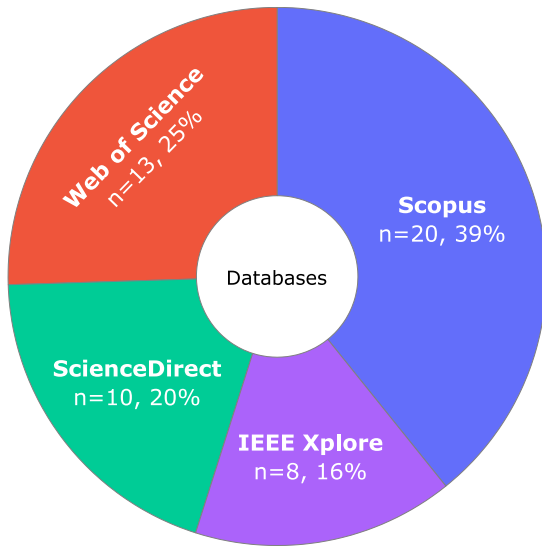


Fig. 11. Distribution of included studies by databases.

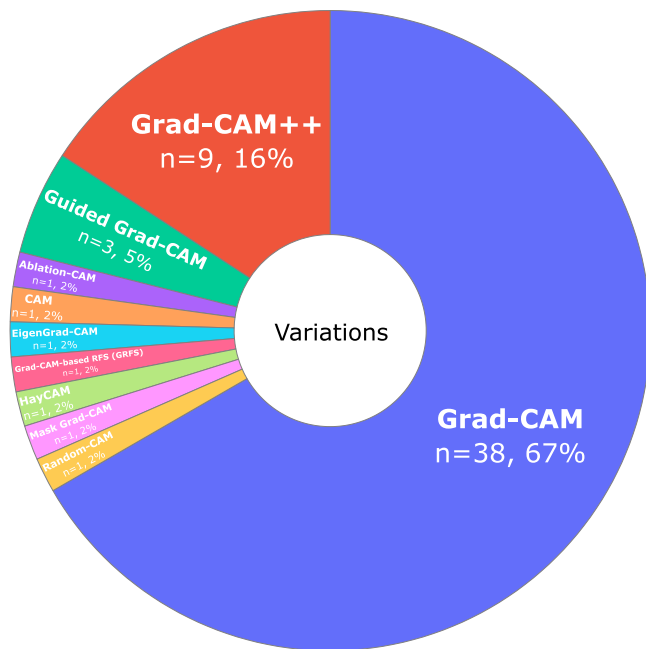


Fig. 12. Variations and enhancements in Grad-CAM.

model's predictions, making it challenging to understand the model's decision-making process. Eigen-CAM mitigates this problem by employing Principal Component Analysis (PCA) to identify and focus on the principal components of the gradient-weighted activation maps. This approach effectively filters out noise, resulting in cleaner and more accurate visual explanations. By concentrating on the most significant regions of the image, Eigen-CAM provides clearer and more interpretable activation maps, which are easier for users to understand. This enhancement in clarity and focus makes Eigen-CAM a powerful tool for building more transparent and trustworthy AI systems, as it offers deeper insights into the workings of deep CNNs. The paper demonstrates the effectiveness of Eigen-CAM in various applications, highlighting its capability to produce more precise and meaningful visual explanations compared to traditional methods. The algorithm based on the Eigen-CAM is explained in Algorithm 5.

5.1.5. HayCAM

HayCAM by Örnek and Ceylan [146] addresses key limitations of traditional Grad-CAM techniques, such as poor localization of relevant regions and susceptibility to noise. Grad-CAM methods are pivotal for visualizing the areas of an image that influence a neural network's decision, but they often highlight broad, less precise areas and can include irrelevant or misleading information due to noise. HayCAM enhances these visual explanations by refining gradient calculations and integrating precise spatial information, resulting in better-localized heatmaps that accurately reflect the features the model uses for predictions. Additionally, advanced filtering techniques in HayCAM reduce noise, providing clearer and more focused explanations, thereby improving interpretability and trustworthiness. Despite its innovative approach, Fig. 12 reveals that HayCAM is among the lesser-used variations of Grad-CAM, representing only 2% of the instances (n=1). The chart shows that the original Grad-CAM is the most prevalent, used in 67% of cases (n=38), followed by Grad-CAM++ at 16% (n=9) and Guided Grad-CAM at 5% (n=3). The distribution highlights the predominance of Grad-CAM and its main variants while acknowledging the existence of diverse adaptations like HayCAM, which contribute to ongoing efforts in improving visual explanation techniques for DL models within the field of XAI. The algorithm based on the HayCAM is explained in Algorithm 6.

5.1.6. Mask Grad-CAM

Mask Grad-CAM by Inbaraj and Jeng [142] is a variant of the Grad-CAM technique designed to provide more precise and interpretable visual explanations by focusing on the most relevant regions of an input image. The original Grad-CAM method, while effective, often produces coarse heatmaps that can include irrelevant background regions, reducing the clarity and interpretability of the visualizations. Mask Grad-CAM addresses this issue by integrating a masking strategy that selectively highlights the most critical areas of the image. This approach involves creating a mask that suppresses the activation of irrelevant regions, thus refining the heatmap to focus only on the important features that contribute to the model's prediction. By doing so, Mask Grad-CAM produces sharper and more accurate localization maps, enhancing the ability to understand and trust the model's decision-making process, especially in scenarios requiring detailed and precise explanations, such as medical imaging or fine-grained object recognition. Mask-Grad-CAM is an advanced framework that combines Mask R-CNN and Grad-CAM to enhance the visual identification and interpretation of DL models. Mask R-CNN is proficient in generating detailed object predictions by loading images as arrays of pixel values and annotating them with class labels, scores, and bounding boxes. This capability is complemented by Grad-CAM, which produces heatmaps that highlight the most relevant regions of an image. However, integrating Grad-CAM with Mask R-CNN introduces challenges such as image pixel issues and quality degradation during heatmap generation. To address these challenges, the Mask-Grad-CAM framework is developed. The algorithm based on the Mask-Grad-CAM is explained in Algorithm 7. This method leverages the strengths of both Grad-CAM++ and Grad-CAM to deliver high-resolution visual representations and fine-grained displays. By doing so, Mask-Grad-CAM provides clearer and more precise visual explanations, making it easier to interpret the model's predictions.

5.2. RQ2: What are the most frequently used ML/DL algorithms used with Grad-CAM by researchers?

From the included literature, Fig. 13 represents the distribution of various ML and DL techniques applied in the context of Grad-CAM. Among the analyzed models, CNNs constitute the most significant share, accounting for 18% of usage, followed by prominent architectures such as ResNet (15%), VGG (14%), and DenseNet (9%). The diversity in model selection illustrates a broad application of architectures, including EfficientNet, GoogleNet, and MobileNet, each contributing

Algorithm 5: Eigen-CAM (Eigen Class Activation Mapping)**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.

Result:

- $L_{\text{Eigen-CAM}}^c$: Eigen class activation map for the target class c .

- 1 **Forward Pass:** Pass the input image I through the CNN model to obtain feature maps A^k at the last convolutional layer.
- 2 **Compute Gradients:** Compute the gradients of the score y^c (corresponding to class c) with respect to the feature maps A^k .
- 3 $\frac{\partial y^c}{\partial A^k} \leftarrow$ Compute gradient
- 4 **Covariance Matrix:** Compute the covariance matrix of the feature maps A^k .
- 5 $C \leftarrow \text{Covariance}(A^k)$
- 6 **Eigen Decomposition:** Perform eigen decomposition on the covariance matrix C to obtain the eigenvectors and eigenvalues.
- 7 $\{v_i, \lambda_i\} \leftarrow \text{Eigen}(C)$
- 8 **Compute Eigen-CAM:** Use the eigenvector corresponding to the largest eigenvalue to weight the feature maps and compute the class activation map.
- 9 $L_{\text{Eigen-CAM}}^c \leftarrow \sum_k v_k A^k$
- 10 **Upsample (Optional):** Optionally, upsample $L_{\text{Eigen-CAM}}^c$ to match the size of the input image I .
- 11 **Return:** The eigen class activation map $L_{\text{Eigen-CAM}}^c$.

Algorithm 6: HayCAM (Hierarchical Attention-based CAM)**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.

Result:

- L_{HayCAM}^c : Hierarchical attention-based class activation map for the target class c .

- 1 **Forward Pass:** Pass the input image I through the CNN model to obtain feature maps A^k at different hierarchical levels (e.g., multiple convolutional layers).
- 2 **Attention Weights:** Compute attention weights α_{kl}^c for each hierarchical level l based on the gradients of the score y^c with respect to the feature maps A^k at that level.
- 3 $\alpha_{kl}^c \leftarrow \text{Attention}(A^k, y^c)$
- 4 **Compute HayCAM:** Compute the weighted sum of the feature maps across all hierarchical levels to obtain the class activation map.
- 5 $L_{\text{HayCAM}}^c \leftarrow \sum_l \sum_k \alpha_{kl}^c A^k$
- 6 **Upsample (Optional):** Optionally, upsample L_{HayCAM}^c to match the size of the input image I .
- 7 **Return:** The hierarchical attention-based class activation map L_{HayCAM}^c .

Algorithm 7: Mask Grad-CAM**Data:**

- I : Input image.
- CNN: Pre-trained CNN model.
- c : Target class.
- M : Mask to be applied to the input image or feature maps.

Result:

- $L_{\text{Mask Grad-CAM}}^c$: Masked class activation map for the target class c .

- 1 **Apply Mask:** Apply the mask M to the input image I or the feature maps A^k at the last convolutional layer. // Or $A^k \leftarrow M \odot A^k$
- 2 $I_M \leftarrow M \odot I$
- 3 **Forward Pass:** Pass the masked input image I_M through the CNN model to obtain feature maps A^k .
- 4 **Compute Class Score:** Compute the score for the target class c , denoted by y^c , before the softmax layer. $y^c \leftarrow \text{score}(c)$
- 5 **Compute Gradients:** Compute the gradients of the score y^c with respect to the feature maps A^k .
- 6 $\frac{\partial y^c}{\partial A^k} \leftarrow$ Compute gradient
- 7 **Global Average Pooling:** Perform global average pooling on the gradients to obtain weights α_k^c .
- 8 $\alpha_k^c \leftarrow \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$ // Where Z is the number of pixels in A^k
- 9 **Compute Masked Class Activation Map:** Compute the weighted sum of the feature maps using the weights α_k^c . Apply ReLU to the result.
- 10 $L_{\text{Mask Grad-CAM}}^c \leftarrow \text{ReLU}(\sum_k \alpha_k^c A^k)$
- 11 **Upsample (Optional):** Optionally, upsample $L_{\text{Mask Grad-CAM}}^c$ to match the size of the input image I .
- 12 **Return:** The masked class activation map $L_{\text{Mask Grad-CAM}}^c$.

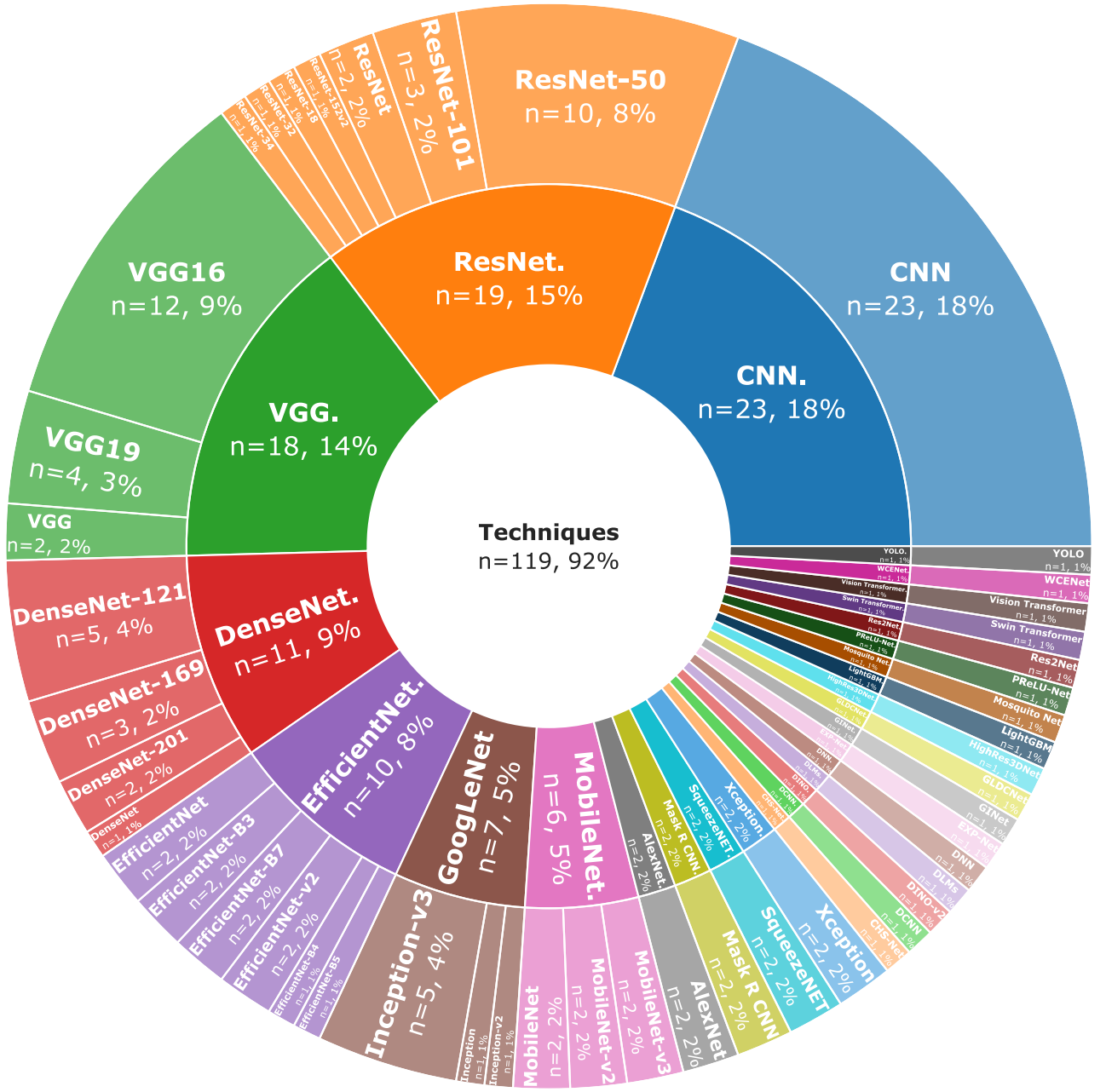


Fig. 13. ML and DL techniques in Grad-CAM.

to a smaller percentage. The presence of specialized and transformer-based models, like Vision Transformers and Swin Transformers, reflects an evolving trend in Grad-CAM applications, leveraging both traditional and advanced architectures to enhance interpretability in various tasks. Brief explanations of these algorithms are given in the coming subsections.

5.2.1. CNN

DL algorithms are supposed to mimic the workings of the human nervous system. These algorithms are models of deep neural networks, which include many hidden layers [62]. CNNs [151] are DL algorithms that can train enormous datasets with millions of parameters, using 2D images as input and convolving them with filters to generate the desired results [152]. CNN models are designed to test their performance on image recognition and detection datasets. The models are pretrained on the ImageNet [153], MNIST [154–156], CIFAR-10, and CIFAR-100 [157] datasets. The mathematical representation of CNN is as

follows:

$$y_{i,j,k} = (x * w)_{i,j,k} + b_k$$

$$= \sum_{m=1}^M \sum_{n=1}^N x_{i+m-1,j+n-1} w_{m,n,k} + b_k \quad (3)$$

where x is the input feature map, w is the filter (or kernel), and b_k is the bias term for the k th output channel. The convolution operation applies the filter over the input to produce the output feature map y .

$$f(x) = \max(0, x) \quad (4)$$

where $f(x)$ is the ReLU (Rectified Linear Unit) activation function, applied element-wise to the input. It outputs the input itself if the input is positive; otherwise, it outputs zero.

$$y_{i,j,k} = \max_{m=1}^M \max_{n=1}^N x_{i+m-1,j+n-1,k} \quad (5)$$

where $y_{i,j,k}$ is the output after applying max pooling over an $M \times N$ window on the input feature map x .

$$y = Wx + b \quad (6)$$

where W is the weight matrix, x is the input vector, and b is the bias vector. The fully connected layer computes the output y as a weighted sum of the input.

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (7)$$

where $\sigma(z)_i$ is the probability of the i th class, z is the input vector to the softmax function, and C is the number of classes.

5.2.2. ResNet

Deeper neural networks are harder to train. A residual learning framework is designed by He et al. [158] to facilitate the training of networks that are far deeper than those previously employed. Rather than learning unreferenced functions, explicitly reformulate the layers to learn residual functions with reference to the layer inputs. Comprehensive empirical data demonstrate that these residual networks are easy to tune and can improve accuracy with much greater depth. On the ImageNet dataset, we analyze residual networks with depths of up to 152 layers, which are 8 times deeper than VGG nets while being less complicated. This work got first place in the ILSVRC 2015 classification problem. Deep residual nets were the foundations of submissions to the ILSVRC and COCO 2015 competitions, where they earned first place in the tasks of ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation. The initial ResNet had 34 layers, but it quickly became more advanced versions, such as ResNet-50 [159], ResNet-101 [160], and ResNet-152 [161], where the numbers represent the total number of layers in the network. These deeper variants of ResNet have exhibited amazing performance on a range of tasks, including picture classification, object identification, and segmentation, making them a cornerstone in computer vision [11].

5.2.3. VGG

The impact of convolutional network depth on its accuracy in large-scale image recognition scenarios. This contribution by Simonyan and Zisserman [162] evaluates networks with increasing depth using very small (3×3) convolution filters. Results show that raising the depth to 16–19 weight layers significantly improves performance compared to previous designs. Two of the best-performing ConvNet models are publically accessible, allowing for additional study into the use of deep visual representations in computer vision. The most well-known variants of VGG are VGG-16 and VGG-19, which are called after the number of weight layers in the network (16 and 19, respectively). These networks employ extremely tiny (3×3) convolution filters, allowing them to learn more detailed aspects of the input pictures while keeping a reasonable amount of parameters. The major novelty of VGG networks is their depth, which allows them to collect hierarchical characteristics, resulting in state-of-the-art performance in picture classification tasks. Despite their computational and memory requirements, VGG networks have become fundamental models in computer vision, acting as benchmarks and pre-trained models for a variety of image processing tasks and transfer learning applications.

5.2.4. DenseNet

The convolutional networks that have shorter connections between layers near the input and those near the output can be trained with far greater depth, accuracy, and efficiency. The author presented the Dense Convolutional Network (DenseNet) [163], a feed-forward network that connects all layers to all other layers. In DenseNet network features $L(L+1)/2$ direct connections, compared to L connections in standard convolutional networks with L layers, which are one link between each layer and the layer after it. Each layer uses the inputs of all the layers that come before it, and all the layers that come after it use

their own feature maps as inputs [68]. DenseNets offer a variety of strong benefits, including resolving the vanishing-gradient issue, enhancing feature propagation, enhancing feature reuse, and significantly lowering the parameter count. The suggested architecture assesses using four very competitive benchmark tasks (CIFAR-10, CIFAR-100, SVHN, and ImageNet) for the identification of objects. For the most part, DenseNets outperform the state-of-the-art while consuming less memory and processing power to reach outstanding performance.

5.2.5. EfficientNet

CNNs are often constructed with a fixed resource budget and then scaled up for improved accuracy when more resources are provided. Tan and Le [164] performs an extensive review of model scaling and concludes that properly balancing network depth, breadth, and resolution can lead to improved performance. Based on this discovery, a novel scaling approach was presented that equally scales all depth/width/resolution dimensions with a simple yet extremely effective complex factor show the efficacy of this strategy on MobileNets and ResNets. To take it a step further, we employ neural architecture search to create a new baseline network and scale it up to produce a family of models known as EfficientNets, which outperform earlier CNNs in terms of accuracy and efficiency.

5.2.6. GoogleNet

Inception is a DL architecture developed by Google researchers [165]. The original Inception network, also known as GoogLeNet, improved image categorization accuracy and efficiency by introducing a unique “Inception module.” This module enables the network to capture multi-scale information by running convolutions with various filter sizes in parallel. The Inception architecture has changed over time, resulting in several versions such as Inception-v2, Inception-v3, Inception-v4, and Inception-ResNet. Each repetition improved accuracy, efficiency, and training speed. Inception-v3, for example, included techniques like factorized convolutions and batch normalization to boost speed even more. Inception-v4 and Inception-ResNet enhanced the benefits of the inception module with residual connections, resulting in significantly greater performance in a variety of computer vision applications.

5.2.7. MobileNet

The author introduced the MobileNets by Howard [166], a family of effective models for embedded and mobile vision applications. The foundation of MobileNets is a simplified architecture that creates lightweight deep neural networks using depthwise separable convolutions. Two straightforward global hyperparameters that effectively balance accuracy and latency were presented by the author. Based on the constraints of the problem, the model builder can select the appropriate size model for their application by adjusting these hyperparameters. In-depth studies on resource and accuracy trade-offs were presented by the author, who also demonstrated strong performance in comparison to other well-known ImageNet classification models. The efficiency of MobileNets in many different use cases and applications, such as large-scale geolocation, facial characteristics, object detection, and fine-grained categorization [167].

5.2.8. Other techniques

AlexNet [168,169] is a DL model that is pre-trained using the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) in 2012. Developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, this CNN was designed to classify images into 1000 categories. It consists of eight layers, including five convolutional layers and three fully connected layers. AlexNet used the Rectified Linear Unit (ReLU) [170] activation function, which improved training efficiency by introducing non-linearity, and dropout regularization, which prevented overfitting by randomly dropping units during training. The model also utilized data augmentation techniques, such as random cropping and mirroring,

to expand the training dataset and enhance the results. Trained on two GPUs in parallel, AlexNet's architecture includes an input layer of $227 \times 227 \times 3$ for RGB images, followed by convolutional layers with varying numbers of filters and strides, and concludes with fully connected layers and a softmax classifier for generating final probabilities. The main objective of deep CNN research has been to increase accuracy. In general, it is possible to find many CNN designs that attain a given accuracy level. Smaller CNN architectures provide at least three benefits with comparable accuracy. In distributed training, smaller CNNs need less communication between servers. Exporting a new model from the cloud requires less bandwidth when using smaller CNNs. On Field Programmable Gate Arrays (FPGAs) and other hardware with constrained memory, smaller CNNs are more practical to implement. The author suggested a compact CNN design known as SqueezeNet [171] to offer each of these benefits. SqueezeNet uses 50 times fewer parameters to achieve AlexNet-level accuracy on ImageNet.

The depthwise separable convolution is followed by direction and ordinary convolution. In this way, an Inception module with the greatest number of layers may be viewed as a depthwise separable convolution. This discovery motivates us to propose a novel deep CNN architecture inspired by Inception, but using depthwise separable convolutions instead of Inception modules. The author demonstrated that the architecture, called Xception [172], performed marginally better than Inception V3 on the ImageNet dataset (the dataset for which Inception V3 was created) and significantly better than Inception V3 on a larger image classification dataset with 350 million images and 17,000 classes. Modern state-of-the-art Computer-Assisted DL algorithms like CNNs, which carry out end-to-end feature extraction and classification, are the foundation of several diagnostic techniques that have made significant contributions to a variety of image identification problems. The author analyzed the effectiveness of a custom-built CNN (Mosquito-Net) designed to distinguish between infected and uninfected blood smears in order to diagnose malaria. Because it has fewer parameters and computational capacity, the CADx system may be deployed on IoT and mobile devices, which makes it extremely advantageous for diagnosis in remote and rural areas lacking medical facilities. Although having ten times fewer parameters and ten times faster inference time than earlier SOTA concepts, statistical analysis shows that the Mosquito-Net [126] provides more reliable results for malaria diagnosis.

5.3. RQ3: What are the most often used performance metrics in the ML/DL methods utilizing Grad-CAM?

Based on the distribution of performance metrics in ML and DL methods that utilize Grad-CAM, the most frequently used performance metrics are shown in Fig. 14, with accuracy leading at 32%, indicating its widespread use for assessing model correctness. Following this, F1-score (14%) and precision (13%) are frequently used, reflecting an emphasis on the balance between recall and precision in model evaluation. Recall itself accounts for 11%, while AUC (8%) and specificity (7%) further demonstrate a focus on distinguishing true positive rates in classification tasks. Sensitivity, training loss, and Cohen's Kappa are also employed, though less commonly, showing a varied approach to performance assessment tailored to specific research goals. Brief explanations of these metrics are given in the coming subsections.

5.3.1. Accuracy

Accuracy is the most commonly used metric, representing the proportion of correctly classified instances among the total instances [173]. It is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

where TP stands for True Positives, TN for True Negatives, FP for False Positives, and FN for False Negatives.

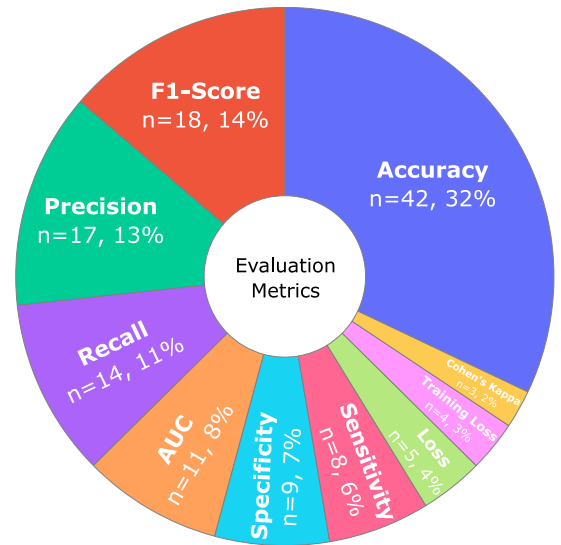


Fig. 14. Most frequently used performance metrics in Grad-CAM research.

5.3.2. Precision

Precision measures the proportion of true positive predictions among all positive predictions [174]. It is a measure of the accuracy of the positive predictions and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

5.3.3. Recall

Recall, also known as Sensitivity or True Positive Rate, measures the proportion of actual positives that are correctly identified by the model [173]. It is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

5.3.4. F1-score

The F1-Score is a harmonic mean of Precision and Recall, providing a balance between the two [173,175]. It is particularly useful when the class distribution is imbalanced. The F1-score is given by:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

where Precision and Recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

5.3.5. AUC

AUC (Area Under the ROC Curve) represents the model's ability to distinguish between classes [176]. It is the probability that a randomly chosen positive instance is ranked higher than a randomly chosen negative instance by the classifier.

$$\text{AUC} = \int_0^1 \text{TPR}(t) d(\text{FPR}(t)) \quad (14)$$

5.3.6. Specificity

Specificity, also known as the True Negative Rate, measures the proportion of actual negatives that are correctly identified [173]. It is defined as:

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (15)$$

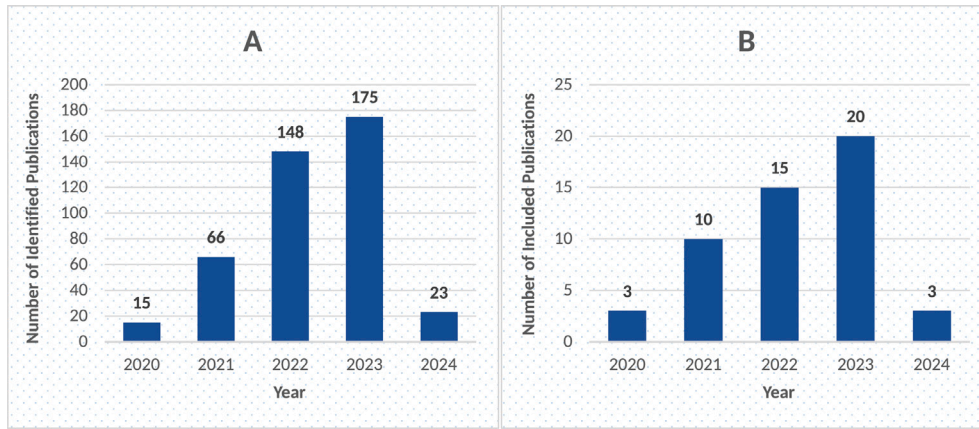


Fig. 15. Publications trend: (A) Number of identified studies and (B) Number of included studies.

5.3.7. Training loss

Training loss is a measure of how well the model is performing during training [177]. It is calculated by comparing the predicted outputs with the actual outputs using a loss function. A common example is the cross-entropy loss [178,179] for classification problems, which is given by:

$$\mathcal{L}_{\text{train}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (16)$$

where: N is the number of training samples, y_i is the true label for sample i , and $\hat{y}_i$ is the predicted probability for sample i .

5.3.8. Prediction speed

Prediction speed refers to the time taken by the model to make predictions on new data [165]. It is often measured in terms of the number of predictions per second (or inference time per sample). The equation for prediction speed S_{pred} can be defined as:

$$S_{\text{pred}} = \frac{N_{\text{pred}}}{T_{\text{pred}}} \quad (17)$$

where: N_{pred} is the number of predictions made, and T_{pred} is the total time taken for making N_{pred} predictions.

5.4. RQ4: What is the publication frequency in the field of the Grad-CAM model?

This sub-section presents the distribution of Grad-CAM-related publications, analyzing yearly trends, publication types, and publishers to uncover prevalent research patterns in the field.

To start with, the yearly trend of Grad-CAM-related publications. Fig. 15 reveals a significant rise in research activity from 2020 to 2024. In the first subplot (A), which shows the number of identified publications, the numbers increased from 15 in 2020 to 66 in 2021, followed by a steep climb to 148 in 2022 and peaking at 175 in 2023. The apparent drop to 23 in 2024 is because the systematic search was conducted earlier, and additional publications are likely yet to be indexed. Similarly, the second subplot (B), which displays the number of included publications, shows a similar rising trend, with 3 included publications in 2020, 10 in 2021, 15 in 2022, and a peak of 20 in 2023, before slightly decreasing to 3 again in 2024. These statistics highlight the increasing scholarly interest and ongoing research efforts focused on Grad-CAM from 2020 to 2024.

Next, the distribution of Grad-CAM-related publications by type, as shown in Fig. 16, highlights the dominance of journal articles and conference proceedings in this field. Journal articles constitute the majority, with 27 publications (53% of the total), followed closely by conference proceedings, which account for 23 publications (45%).

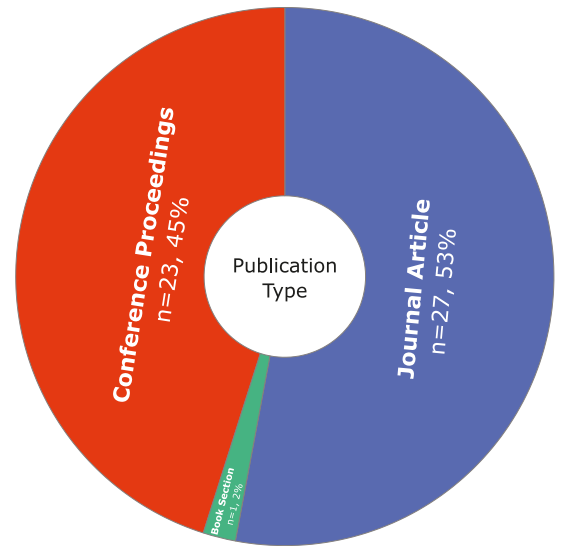


Fig. 16. Distribution of included studies by publication type.

Additionally, there is a small contribution from book sections, with only 1 publication (2%).

Finally, the distribution of publications by the publishers, as presented in Fig. 17, shows a concentration of research published by a few key publishers. The leading publisher is *IEEE*, with 22 publications, followed by *Elsevier* with 11 publications. Then, *Springer* and *MDPI* contributed 6 and 5 publications, respectively. Other publishers, such as *SPIE*, contributed 2 publications, while *CRC Press*, *Wiley*, *Maruzen*, *PeerJ*, and *IJETA* each contributed 1 publication. This distribution indicates that Grad-CAM research is primarily disseminated through major academic publishers, with *IEEE* and *Elsevier* playing dominant roles.

6. Conclusion

This SLR is concluded based on its theoretical implications, limitations, and closing remarks in the Sections Section 6.1, 6.2, and 6.3, respectively.

6.1. Theoretical implications

In this paper, a comprehensive SLR was conducted to examine the role and evolution of Grad-CAM as an essential tool in XAI, particularly within the realms of ML and DL. The primary objective was to investigate the diverse applications of Grad-CAM and its variations,

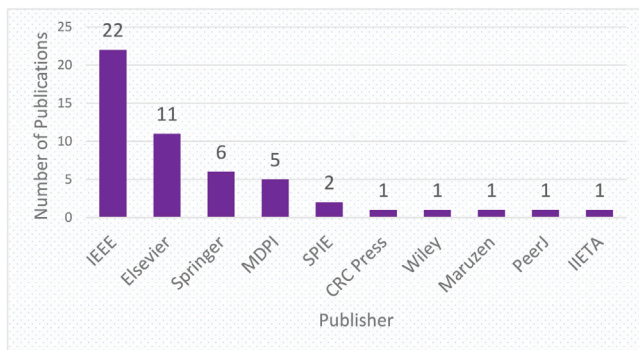


Fig. 17. Distribution of included studies by publisher.

and to assess the trends in recent research spanning from 2020 to 2024. By exploring a wide range of databases: Scopus, Web of Science, IEEE Xplore, and ScienceDirect, 427 manuscripts were identified and 51 were selected for in-depth analysis based on relevance and contribution to the field. The findings underscore the growing recognition of XAI techniques like Grad-CAM as indispensable for enhancing model interpretability and fostering trust in ML/DL applications. Grad-CAM, with its ability to visualize CNN decision-making processes by highlighting significant image regions, has proven particularly valuable across numerous domains, including healthcare, autonomous systems, and image classification. The increasing frequency of publications within the reviewed timeframe suggests an upward trend in the adoption and refinement of Grad-CAM, as researchers seek to enhance the interpretability and transparency of complex models.

The analysis also revealed a diverse range of Grad-CAM variations, each tailored to address specific challenges associated with traditional CNN interpretability. These variations aim to improve localization accuracy, increase computational efficiency, and extend applicability to a broader array of DL architectures beyond CNNs. Such innovations indicate that Grad-CAM is not only a relevant tool in current ML/DL research but is also continuously evolving to meet the dynamic needs of the field. This SLR highlights Grad-CAM as a vital component of the XAI landscape, bridging the gap between complex DL models and user interpretability. Future research may benefit from further refining Grad-CAM methods to address limitations in computational cost and accuracy while expanding its application to other areas of AI. Additionally, exploring hybrid approaches that integrate Grad-CAM with other XAI techniques could provide a more holistic understanding of model behaviors, thereby enhancing transparency and trust in AI-driven decision-making processes across various sectors.

6.2. Limitations

This SLR focuses on the use of Grad-CAM to support interpretability in medical imaging applications. It covers research published from 2020 to 2024, meaning that earlier foundational studies or initial explorations of visual explanation methods may not be included. Nonetheless, the aim is to present a timely and focused summary of the most recent advancements in Grad-CAM usage across the medical imaging field. One key limitation of this review is its concentration on only the medical imaging domains, which were selected based on their relevance and the availability of substantial research. While this choice allows for more detailed analysis within each domain, it may have excluded insights from other areas where Grad-CAM is also being explored. The paper selection process was guided by a systematic screening strategy, as outlined in Table 3, ultimately narrowing 427 initially identified papers down to 51 that met the review standards. This review also considered only English language articles and focused on literature from the disciplines of Computer Science, Engineering,

and Medical Imaging. As a result, studies published in other languages or in different academic fields may not have been captured in the search. Lastly, while the review categorizes a variety of Grad-CAM implementations and enhancements, it does not include a comparative experimental evaluation across standard datasets or tasks. Incorporating such benchmarking could provide further practical value for researchers looking to apply these methods. Despite these constraints, this SLR offers meaningful insights into how Grad-CAM is currently being used in medical imaging and serves as a useful resource for those developing interpretable DL models in healthcare contexts.

6.3. Closing remarks

This SLR followed a structured four-phase screening process aligned with the PRISMA methodology as illustrated in Fig. 10. The included studies were thoroughly reviewed and analyzed from multiple perspectives to address the four formulated research questions (RQ1, RQ2, RQ3, and RQ4).

For RQ1, this review covered a range of Grad-CAM variations introduced to enhance its adaptability and effectiveness across medical imaging tasks. While the original Grad-CAM remains the most widely used (67%), several enhancements have emerged, such as Grad-CAM++ (16%) and Guided Grad-CAM (5%), along with other less common but notable adaptations (Fig. 12). These variants aim to improve feature localization, refine visual explanations, or better align with specific model architectures and imaging requirements. The presence of multiple enhancements reflects ongoing efforts to tailor Grad-CAM for different interpretability needs in clinical and research settings. By analyzing these variants, this review provides a clearer understanding of how Grad-CAM continues to evolve, offering researchers valuable insights for selecting or adapting the most suitable approach in their own applications.

For RQ2, the analysis of the selected studies revealed that CNNs are by far the most widely used DL architecture in conjunction with Grad-CAM (Fig. 13). CNNs account for the largest share of usage (18%), followed by ResNet (15%), VGG (14%), and DenseNet (9%). These architectures were frequently applied across various medical imaging tasks. A few studies also explored the use of more recent architectures, including transformer-based models, though these were less common. The findings indicate that Grad-CAM is most compatible with models that have clear feature hierarchies and spatial representations, which explains its strong integration with CNN-based approaches. These insights help clarify which types of models benefit most from Grad-CAM's visual explanations and point to potential areas where its application could expand as new DL methods emerge.

For RQ3, the analysis showed that studies using Grad-CAM alongside ML/DL models commonly report metrics such as accuracy, precision, recall, F1-score, and AUC-ROC (Fig. 14). These metrics are primarily used to evaluate the classification performance of the models prior to applying Grad-CAM for interpretability purposes. Their consistent use highlights the importance of validating the model's predictive capability as a foundation for generating trustworthy explanations. Additionally, some studies also employed other metrics like specificity, sensitivity, training loss, and Cohen's Kappa, reflecting a broader range of evaluation strategies tailored to the research context. By identifying these commonly used performance measures, this review provides a helpful reference for selecting appropriate evaluation criteria in future studies involving Grad-CAM.

For RQ4, the review examined the publication trends of Grad-CAM-related research in medical imaging by analyzing the distribution of studies across different publishers, publication types (Fig. 17), and years (Fig. 15). The findings indicate a noticeable increase in publications between 2020 and 2024, reflecting growing interest in XAI and the integration of interpretability tools like Grad-CAM in healthcare.

Most of the selected studies were published in peer-reviewed journals, followed by conference proceedings, highlighting both academic rigor and ongoing innovation in the field. Additionally, a diverse range of publishers was identified, suggesting widespread recognition of Grad-CAM's relevance across multiple research platforms. These patterns point to a steadily rising focus on explainability in DL, especially in medical contexts where transparency is critical.

CRedit authorship contribution statement

Abdul Muiz Fayyaz: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Said Jadid Abdulkadir:** Writing – review & editing, Writing – original draft, Resources, Project administration, Formal analysis, Conceptualization. **Noureen Talpur:** Writing – review & editing, Project administration. **Safwan Mahmood Al-Selwi:** Writing – review & editing, Visualization, Methodology, Formal analysis. **Shahab Ul Hassan:** Writing – review & editing, Formal analysis. **Ebrahim Hamid Sumiea:** Writing – review & editing, Formal analysis.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used different AI tools to improve the language and readability, then reviewed and edited the content, taking full responsibility for the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the Universiti Teknologi PETRONAS under Yayasan Universiti Teknologi PETRONAS (YUTP-FRG/015LC0-521) and (YUTP-FRG/015LC0-578) for providing the required facilities to conduct this research work.

Data availability

The data used to conduct this SLR is available at: <https://github.com/abdulmuizfayyaz/GradCAM-SLR>.

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